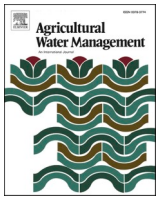
Agricultural Water Management 333 (2026) 110583



Contents lists available at ScienceDirect

Agricultural Water Management

journal homepage: www.elsevier.com/locate/agwat





Breaking biological and abiotic limits of agricultural water use efficiency – A review

Xianzhi Deng [a], Liangsheng Shi [a,*], Shaozhong Kang [b], Zhiying Liu [a], Haolin Xu [a], Yujie Wang [a], Zailin Huo [b], Yunkai Li [b], Jinmin Li [a], Tinghan Wang [a]

[a] State Key Laboratory of Water Resources Engineering and Management, Wuhan University, Wuhan, China
[b] State Key Laboratory of Efficient Utilization of Agricultural Water Resources, China Agricultural University, Beijing, China

ARTICLE INFO

Keywords:
Water use efficiency
Agricultural water use
Water saving limits
Smart irrigation
Multi-scale regulation

ABSTRACT

Water use efficiency (WUE) is a multidimensional concept spanning genetic, agronomic, and hydrological scales, yet the lack of an integrated framework linking its biological and abiotic determinants continues to limit progress toward agricultural water saving. We propose a paradigm shift in agricultural WUE through dual-breakthrough: overcoming the biological limits to crop water use and addressing the abiotic limits on water delivery. We methodically review the definitions and developmental trajectory of agricultural WUE at micro, meso and macro spatial scales. We highlight that gene editing, synthetic biology, and nanotechnology will enhance water use efficiency in crops, but understanding system-level gene regulatory networks is needed to explore how micro-level interventions regulate plant traits. Overcoming abiotic constraints requires innovations in irrigation methods and equipment, as well as strategies for irrigation and water delivery. Improved water savings relies on mechanistic elucidation of interactions in the microbe-soil-crop-aquifer-atmosphere system, necessitating advanced big data processing, intelligent modelling, and adaptive water use decisions within digital entities. The ongoing digital transformation of biological systems, agricultural lands, and watersheds into high-fidelity digital representations means artificial intelligence will play an essential role in sustainable water management within complex ecosystems. Therefore, the primary challenge ahead lies in unlocking the complexities of biological systems and real-world environments to integrate the mechanisms that govern water use efficiency across multiple scales.

1. Introduction

Agriculture serves as the cornerstone for the majority of economies globally, driving significant contributions to Gross Domestic Product (GDP) while simultaneously ensuring food security (Tom et al., 2024). However, agricultural water use is a critical component of global water consumption, accounting for approximately 70% of total usage worldwide (Boretti and Rosa, 2019). Projections indicate that by 2050, agricultural water demand could surge by 20–30% (Preite et al., 2024), driven by population growth and the need to increase food production by 70% to meet global demands (Velasco-Muñoz et al., 2018). This alarming trend underscores the severe pressure on freshwater resources. Without significant improvements in water use and sustainable practices, we risk exacerbating water scarcity, which could adversely impact food security and ecosystems globally. Therefore, urgent action is required to balance agricultural productivity with water stewardship (Deng et al., 2006).

Recognizing the critical link between agricultural output and water availability, particularly under pressures from shifting distribution patterns of water resources and increasing demands, the scientific community is prioritizing efficient water utilization (Preite et al., 2023), striving for “more crop per drop” (Geilfus et al., 2024). While agricultural water saving ultimately aims to reduce actual water withdrawals or consumptive water use at field, basin, or regional scales, it is frequently pursued through improvements in water use efficiency (WUE), which describes the amount of agricultural output obtained per unit of water used. As a result, WUE has emerged as a central metric in efforts to build resilient and sustainable food systems (Kilemo, 2022).

WUE is a multifaceted indicator that quantifies the effectiveness of water utilization in agricultural systems, encompassing a chain of efficiencies including hydrological, physiological, conveyance/engineering, and application effectiveness (Kilemo, 2022, 2022). Central to its

* Corresponding author.
E-mail address: liangshs@whu.edu.cn (L. Shi).

https://doi.org/10.1016/j.agwat.2026.110583
Received 13 January 2026; Received in revised form 26 June 2026; Accepted 26 June 2026
Available online 29 June 2026
0378-3774/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

definition is the core question: how much of the supplied water (from irrigation or rainfall) is effectively absorbed and transpired by plants, ultimately contributing to biomass production (Seckler et al., 2003)? The complexity in defining WUE arises from differing disciplinary perspectives (Sadras et al., 2018). From a biological perspective, WUE is commonly expressed as biomass or yield produced per unit of transpired or evapotranspired water, reflecting intrinsic physiological and genetic constraints associated with photosynthesis, stomatal regulation, and root water uptake (A. G. Condon et al., 2002; Medrano et al., 2010; Ruggiero et al., 2017; Viets, 2015). From an abiotic or engineering perspective, WUE-related evaluations focus on irrigation efficiency, water conveyance performance, and field-scale water distribution, emphasizing technological and infrastructural constraints that govern water losses during transport and application (Kilemo, 2022, 2022; Seckler et al., 2003).

Operating across diverse spatial and temporal scales, WUE integrates physiological, agronomic, engineering, and management processes (Clothier et al., 2020; Fernández et al., 2020; Leakey et al., 2019). Its scope spans from the molecular and cellular levels to organs, plants, populations, fields, and entire watersheds, encompassing temporal resolutions from instantaneous to hourly, daily, monthly, seasonal, and annual periods (Condon, 2020). The very nature of WUE's multifaceted influence across these scales means it is a profoundly complex attribute, intricately influenced by a confluence of interwoven factors. These span from fundamental crop physiological characteristics, such as root architecture, stomatal conductance, canopy structure, and phenological development, to dynamic natural environmental conditions, including temperature, humidity, CO_2 concentration, solar radiation, and highly variable precipitation patterns (Vadez et al., 2023). Furthermore, sophisticated management strategies, encompassing irrigation methods and scheduling, nutrient management, soil health practices, and planting densities, play an equally crucial role, though their effective implementation is often mediated by societal factors such as local knowledge, resource constraints, and collective decision-making frameworks (Hoover et al., 2023). Research consistently demonstrates that WUE distribution patterns are not governed by isolated variables but rather emerge from the dynamic and synergistic interplay of these diverse elements - encompassing climate regimes, diverse soil properties, and specific biological traits of crops (Dou et al., 2024). Understanding these intricate, multi-layered relationships and their feedback mechanisms is fundamental not only to precisely quantify WUE but, more importantly, to developing truly effective, context-specific, and holistic strategies toward agricultural water saving.

A targeted search of the peer-reviewed literature was undertaken to thoroughly summarize the classic and emerging articles pertinent to this review. This focused approach involved consulting databases such as Google Scholar, Web of Science, PubMed, and Scopus to identify and select key, high-quality articles on the topic. Articles were selected based on the key terms "water use efficiency", "agriculture water use", and "water saving" combined with Boolean logic operators. Our literature search focused on English and Chinese publications up to January 1st, 2026, and grey literature (e.g., thesis, dissertations, reports) and duplicates were excluded. To be included in the review, each article had to meet specific criteria: first, incorporating terms related to agricultural water use and efficiency, such as "water saving", "water use", "water distribution", or "water use efficiency", immediately followed by agriculture-related terms like "agriculture", "crop", "field", "watershed", or "irrigation" within their title, keywords, or abstract; and second, studies had to include discussions related to agricultural water use across disciplines such as natural science, hydrology, agriculture, engineering, or relevant social science aspects. To incorporate the latest advancements in the field, papers from the last decade were given priority, while classic literature was also included to summarize historical key findings. Given the heterogeneity of WUE definitions, crop types, spatial scales, and methodological approaches, this review was designed as an integrative synthesis rather than a formal meta-analysis.

The research summary showed that current strategies to enhance WUE are indeed being pursued vigorously across these distinct yet interconnected scales, reflecting the recognized need for a comprehensive approach. At the most fundamental level, the crop physiology scale focuses on molecular regulation of drought-resistant crop varieties and optimization of water metabolism pathways (Z. He et al., 2024). This includes research into aquaporins, stomatal regulation, root exudates, and the genetic markers associated with improved water uptake and retention within the plant. Moving up to the field management scale, efforts center on implementing precision irrigation techniques, strategic water-fertilizer integration to minimize nutrient leaching and maximize uptake, and various soil improvement practices aimed at enhancing soil water holding capacity and reducing evaporation (Lu et al., 2022). Finally, at the largest, irrigation district scale, strategies involve the development of dynamic water distribution models that utilize real-time data and forecasting to optimize supply, along with advanced systems for precise irrigation decisions that ensure equitable and efficient water delivery across large agricultural areas. This also encompasses modernization of infrastructure, inter-basin transfers, and demand-side management policies (Liang et al., 2020), as well as socio-economic factors influencing collective decision-making on water allocation (Visser et al., 2024; Qinhong Xu et al., 2024). Each of these scales offers unique opportunities for WUE improvement.

Although numerous review articles have addressed evaluation metrics and improvement strategies for agricultural water use efficiency (Antu et al., 2024; Bwambale et al., 2022; Condon, 2020; Singh et al., 2010; Wallace, 2000), most of them focus on specific indicators or single spatial scales, lacking a comprehensive understanding of the "gene--plant-field-irrigation district" multi-scale system (Fig. 1). This limitation is compounded by the reality that progress within each individual scale is also impeded by the sheer complexity and intricate interdependencies of factors influencing WUE (Burridge et al., 2022). Recognizing that isolated improvements at a single level - whether it's a breakthrough in plant genetics or a new irrigation scheduling algorithm - are inherently insufficient to fully address the overarching challenge of water scarcity, our efforts must span a continuum. This continuum extends from the microscopic genetic and cellular level within the plant to the macroscopic, landscape-level management of entire irrigation districts, critically acknowledging the pervasive feedback loops and dependencies between these scales (Condon, 2020). For instance, genetic improvements in drought tolerance may not fully manifest their potential without complementary advances in field-level precision irrigation and optimized nutrient delivery (Sadras and Richards, 2014). However, the effective integration of these multi-scale strategies is also shaped by societal and rural dynamics, such as collective decision-making processes, water governance structures, and local adoption barriers (Visser et al., 2024; Qinhong Xu et al., 2024). Conversely, highly efficient irrigation infrastructure at the district scale cannot entirely compensate for crop varieties with inherently low water-use efficiency or suboptimal field management practices (Sadras and Richards, 2014). This multi-scale perspective is therefore not merely advantageous but absolutely crucial for developing comprehensive, resilient, and truly effective water-saving strategies that can withstand diverse environmental conditions, adapt to local operational realities, and deliver tangible, sustainable gains in agricultural productivity (Dunn et al., 2019; Kang et al., 2017; Maurel and Nacry, 2020; Papacek et al., 2019).

This review differs by explicitly integrating biological and abiotic perspectives across multiple scales, from gene and plant processes to field and irrigation district management, with the aim of providing a unified conceptual framework for understanding and improving WUE. Our core objectives are to integrate transformative technological advances across biological, agronomic, and systemic domains, effectively connecting microscale (gene-plant), mesoscale (field), and macroscale (irrigation district) levels. We propose a paradigm shift to enhance agricultural WUE through a dual breakthrough: overcoming the biological limits of crop water use and addressing the abiotic limits of water

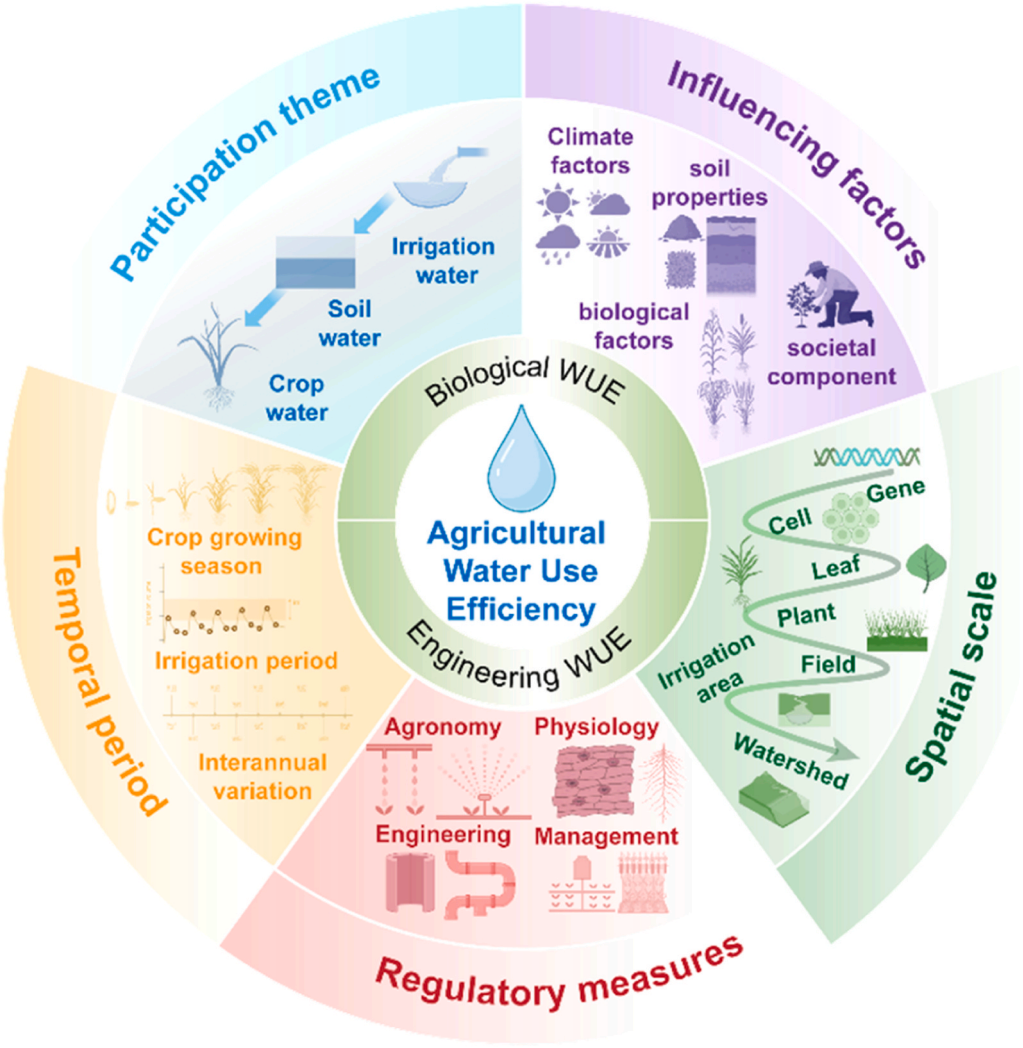


Fig. 1. | Overview of water use efficiency in agriculture. This figure comprehensively outlines agricultural water use efficiency, including its components, influencing factors, temporal and spatial scales, and integrated management strategies.

delivery. Biological limits of crop water use refer to intrinsic plant physiological and genetic constraints, such as photosynthetic efficiency, root water uptake capacity, and stomatal regulation. Meanwhile, abiotic limits of water delivery encompass systemic barriers beyond plant biology, including field-level irrigation infrastructure, soil hydraulic properties, climate variability, basin-scale water allocation logistics, and socioeconomic policies governing water resource accessibility and equity. We will systematically elaborate on the definition, fundamental characteristics, limiting factors, and the development of improvement measures at each level of agricultural WUE, and based on this, propose future directions. Specifically, we will decode water flows from the fundamental soil-plant-atmosphere continuum to regional hydrological cycles, identifying critical interventions in physiological adaptation, precision irrigation, and precision water delivery. Ultimately, this review seeks to provide a consolidated perspective on how breakthroughs, ranging from synthetic biology reengineering plant hydraulics to irrigation optimization and dynamic water delivery decision-making, can collectively drive a paradigm shift in WUE optimization. By bridging the knowledge gaps between different levels of organization, we offer a holistic view of WUE challenges and opportunities. This integrated perspective, coupled with the accelerating pace of discovery, positions us at the forefront of a paradigm shift in agricultural water management. We are at the dawn of a potential breakthrough in water use efficiency, where technological advancements could redefine how water is managed in the global agricultural landscape.

2. Calculation and current fundamentals of agricultural water use efficiency across different scales

Water use efficiency (WUE) is not a single measurable indicator but a broad conceptual framework encompassing a suite of quantitative metrics that describe the relationship between agricultural output and water use (Kilemo, 2022, 2022). These metrics differ in their definitions, calculation methods, and evaluation purposes depending on the scale of analysis, the object of assessment, and the management objectives (Hatfield and Dold, 2019; Sparks, 2012). In irrigated agriculture, WUE-related indicators are commonly categorized into physiological efficiency at the crop scale, irrigation efficiency at the field scale, and water conveyance or delivery efficiency at the irrigation district scale (Howell, 2001). The following sections clarify the definitions, calculation logic, and applicability of these indicators across spatial scales.

Agricultural irrigation aims to optimize crop yield by supplementing farmland water to meet crop needs. This process, which spans from water source to economic yield, involves several key stages where WUE-related indicators are critical for assessing water losses and utilization efficiency and ideally includes three main phases. First, water distribution engineering transports water via channels or pipes, a substantial portion of water is lost through evaporation and seepage (Mutema and

Dhavu, 2022; Paltineanu et al., 2013; Serafeim et al., 2024). Second, the application of irrigation water, whether by sprinkler, micro-irrigation, or surface methods, is often associated with losses such as uneven distribution, surface runoff, and deep percolation (Yousaf et al., 2021). Third, crops absorb water to meet the demand of transpiration—a key process that drives the physiological activities essential for growth and yield formation (Barna et al., 2018). Through these complex processes, crops generate economic yield. While transpiration is essential for crop growth and yield formation, there exists a point beyond which further increases in transpiration do not proportionally increase yield but lead to significant water wastage (Steduto et al., 2012). Water wastage can occur at every stage—from transport and irrigation to soil storage and crop utilization (Sinha Roy, 2023) – making it a significant challenge to improve water use efficiency-related performance across different stages and scales (Fig. 2).

Importantly, improvements in WUE-related indicators at different stages or scales do not necessarily translate into actual agricultural water saving (Perry et al., 2009). While WUE metrics quantify how efficiently water is converted into biomass or yield within specific components of the agricultural system, real water saving refers to a measurable reduction in water withdrawals or consumptive use at the plant, field, irrigation district, or basin scale (Grafton et al., 2018; Ward and Pulido-Velazquez, 2008). In practice, efficiency gains achieved at one stage of the water-use chain may improve productivity without reducing total water use, and in some cases may even increase overall water consumption through expanded irrigated area or intensified production (Kilemo, 2022). Therefore, establishing a linkage between multi-scale WUE improvements and actual agricultural water saving requires explicit consideration of system boundaries, scale effects, and water allocation constraints (Kilemo, 2022). Within this framework, agricultural water resources are understood as the available freshwater stocks accessible to agricultural systems, while agricultural water use refers to the portion of these resources that is withdrawn and consumed through evapotranspiration (Falkenmark and Rockström, 2006). Efficiency indicators at different levels serve as diagnostic tools for identifying losses and performance within the agricultural water system, rather than as direct measures of agricultural water saving.

WUE-related indicators are defined differently depending on the scale of observation (Sparks, 2012). At the microscale, it pertains to crop physiological efficiency, measured as the yield produced per unit of water consumed (Section 2.1). This is determined by species traits (Zahoor et al., 2019). At the mesoscale and macroscale, it quantifies hydrological performance, such as irrigation system efficiency (Section 2.2, Section 2.3). This is driven by engineering interventions (Stewart and Howell, 2003). Multi-scale optimization requires the integration of biological adaptation, technological upgrades, and systemic management (Fig. 2).

2.1. Microscale perspective on crop water use efficiency

At the microscale, crop water use efficiency is governed by physiological processes controlling water-carbon exchange at the plant level, with leaf-scale processes constituting a fundamental component of plant water use efficiency rather than an independent evaluation scale. Physiological water use efficiency in crops measures the relationship between crop yield and water consumption. In agronomic practice, crop-level physiological WUE is often operationally expressed using yield-based indicators such as the water consumption coefficient (K_w), which indicates the amount of water needed to produce a unit of yield; a higher K_w signifies lower water use efficiency (Hongren et al., 2008).

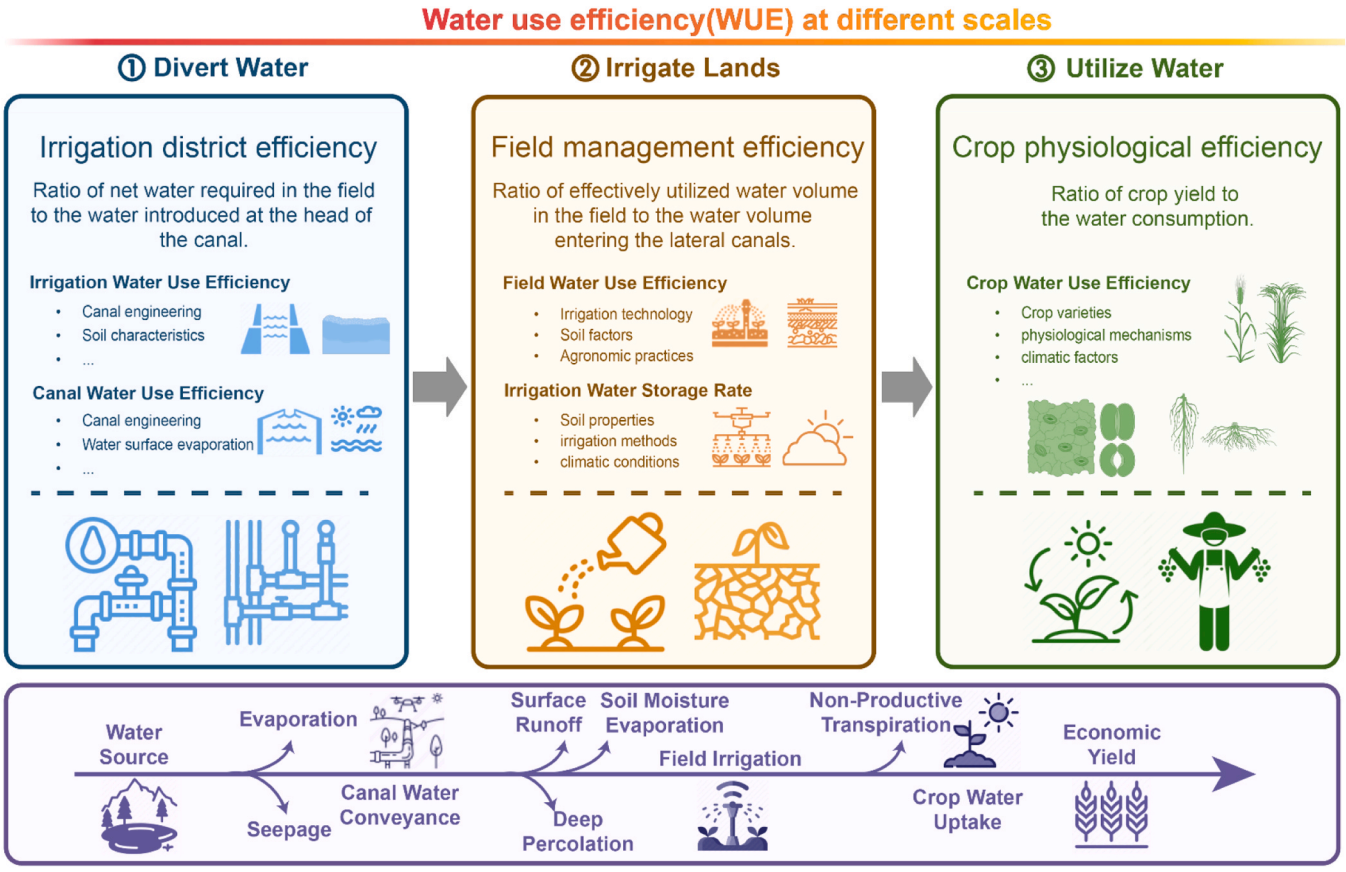


Fig. 2. | Definitions of water use efficiency at different scales. Definitions of water use efficiency exist across microscale crop water use, mesoscale field irrigation, and macroscale irrigation district water conveyance. The pathway from water sources to economic yield involves critical stages including water conveyance, irrigation application, and crop water utilization, with water losses occurring at each stage.

Water use efficiency has gained prominence as it quantifies yield per unit of water consumed, allowing for straightforward comparisons across different crops or conditions.

At the leaf scale, as part of plant-scale physiological regulation, water use efficiency can further be divided into intrinsic transpiration efficiency and instantaneous water use efficiency (Vadez et al., 2023). The calculation formula is as follows:

$$WUE_{int} = \frac{A}{G_s} = \frac{G_{sC} \times (C_a - C_i)}{G_{sW} \times (W_i - W_a)} \quad (1)$$

$$WUE_{int} = 0.6C_a \times \frac{(1 - C_i/C_a)}{(W_i - W_a)} \quad (2)$$

$$WUE_{ins} = \frac{A}{T} \quad (3)$$

WUE_{int}, or "intrinsic transpiration efficiency," is defined as A/G_s, where A is the CO_2 assimilation rate and G_s is stomatal conductance. It can also be expressed as A/T, with T representing the transpiration rate. G_{sC} and G_{sW} denote stomatal conductance for CO_2 and water, respectively, while C_a and C_i are the CO_2 concentrations in the air and stomatal chamber, and W_a and W_i are the vapor pressures in the air and stomatal chamber. Eq. 2 simplifies Eq. 1, assuming a G_{sC} to G_{sW} ratio of approximately 0.6. WUE_{ins} refers to instantaneous water use efficiency, calculated as A/T, where T is the transpiration rate.

$$WUE_y = \frac{Y}{WU} \quad (4)$$

At the plant scale, crop water use efficiency is commonly quantified using yield-based indicators, defined as the yield per unit of water consumption, where Y represents the crop yield, which can be expressed as total biomass (Y_b) or economic yield (Y_e); WU is the crop water use. The term can refer to water consumed by the crop through evapotranspiration, or to water delivered to the crop by irrigation and rainfall, or a combination of both.

The crop water use efficiency changes with scale as (Hoover et al., 2023) :

$$\underset{g_s}{A} \rightarrow \underset{E}{A} \rightarrow \underset{Water\ loss}{Biomass} \rightarrow \underset{ET}{Yield} \rightarrow \underset{ET}{GPP} \quad (5)$$

Where gross primary production (GPP) is the total carbon fixed by plant photosynthesis, while evapotranspiration (ET) is the combined water loss from soil evaporation and plant transpiration.

There are significant variations in water use efficiency at the plant scale among different crops (Fig. 3). Studies have shown that wheat WUE_y ranges from 0.54 to 1.73 kg/m^3 , rice from 0.46 to 1.84 kg/m^3 , and maize from 0.65 to 11.81 kg/m^3 (Foley et al., 2020; Zwart and Bastiaanssen, 2004). While many C_3 cereal and legume crops show WUEy values mostly below 3 kg m^{-3} , certain high-yield C_3 crops such as potato and tomato can reach substantially higher WUEy under intensive management. In contrast, C_4 crops generally maintain higher intrinsic efficiency and higher upper-bound WUEy values. Reported values reveal that intrinsic transpiration efficiency WUE_{int} varies from 5 to 324 μmol CO_2 mol^{-1} H_2O, and instantaneous water use efficiency WUE_{ins} ranges from 2 to 6 μmol CO_2 mol^{-1} H_2O (Gago et al., 2014; Pou et al., 2012).

Crop water use efficiency is fundamentally determined by inherent genetic and physiological characteristics, including species-specific characteristics, varietal differences in drought tolerance, stomatal regulation, cuticle properties (Tambussi et al., 2007), and the balance between transpiration and biomass production (Morison et al., 2008). Beyond these intrinsic factors, WUE is significantly modulated by external influences such as climatic conditions like precipitation variability and temperature regimes (Abbate et al., 2004), agricultural practices like optimized irrigation and soil management (Anthony G. Condon et al., 2002), and soil properties including water-holding capacity and nutrient availability (Hatfield et al., 2001). Furthermore, environmental stressors such as drought, salinity, and nutrient deficiency can severely reduce WUE by limiting water uptake and disrupting metabolic processes (Blum, 2009). Given the complex interplay of these factors, understanding objective and reasonable methods for evaluating crop water use efficiency, and advancing measures to improve it, remains critically important.

Several methods exist to measure crop water use efficiency, including direct measurement (Ebmeyer and Hoffmann, 2022), gas exchange (Pan et al., 2022), stable isotope analysis (Munjonji and Ayisi, 2021), pot weighting (Alataway et al., 2019), and eddy covariance (Jiang et al., 2020). Direct measurement calculates WUE based on observed water consumption and crop yield, often using energy and water balance coupling (Ebmeyer and Hoffmann, 2022). Gas exchange is limited to leaf-scale measurements, while stable isotope technology provides high sensitivity and accuracy for assessing both WUE and water sources (Pan et al., 2022). Pot weighting is restricted to potted plants (Alataway et al., 2019), and eddy covariance allows for WUE measurement at larger scales (Jiang et al., 2020). However, many of these methods are labor-intensive, require destructive sampling, and have limited spatial coverage, making continuous observation challenging and prone to error.

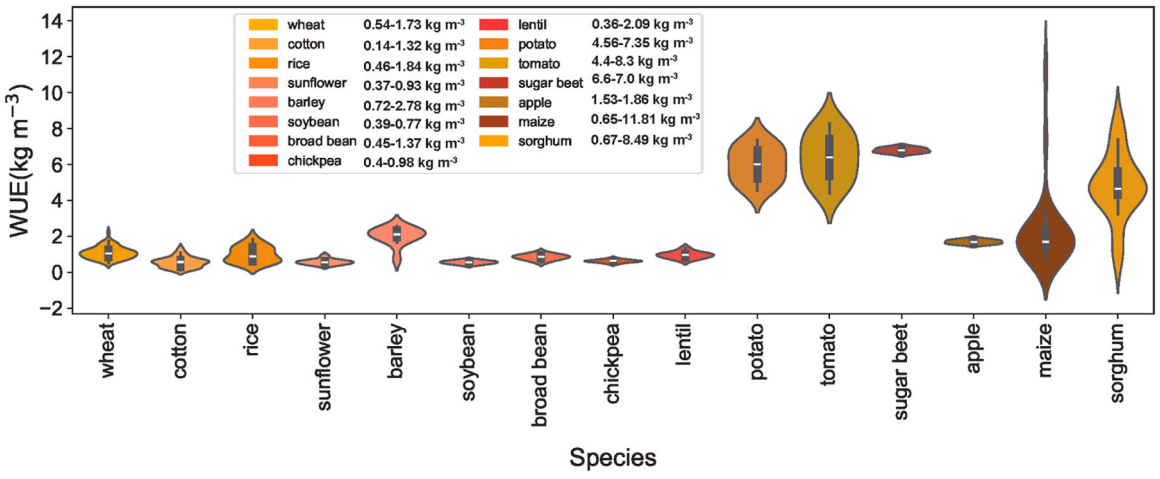


Fig. 3. | The ranges of water use efficiency at the plant scale for various crops. Data are compiled from over 160 publications, with specific data provided in the appendix.

2.2. Mesoscale perspective on field irrigation efficiency

At the field scale, WUE-related performance is commonly quantified using irrigation efficiency indicators, defined as the ratio of the effectively utilized water in the field to the total amount of water entering the main canal, reflecting the effective utilization of irrigation water in the field (Stewart and Howell, 2003). It encompasses various aspects, including reducing deep percolation and surface runoff during irrigation, as well as minimizing soil water evaporation losses through moisture conservation.

The formula for calculating the field water use coefficient or field irrigation efficiency is as follows :

$$\eta_{field} = \frac{A_{agri} \cdot m_{net}}{W_{agri\ net}} \tag{6}$$

A_{agri} represents the irrigated area of the farm canal (ha); m_{net} refers to the net irrigation quota or effective irrigation quota. This excludes deep percolation and surface runoff for dryland farming (m^3/ha); $W_{agri\ net}$ is the net water volume released by the farm canal (m^3). This utilization coefficient is an important WUE-related indicator reflecting the field engineering conditions, irrigation techniques, irrigation methods, and management level (Lu et al., 2022).

Irrigation water storage rate refers to the percentage of the actual irrigation water depth stored in the planned wetting layer after irrigation relative to the maximum storable irrigation water depth in the planned wetting layer. It can be calculated using the formula:

$$E_s = \frac{Z_s}{Z_q} \tag{7}$$

Where E_s is irrigation water storage rate, Z_s is the irrigation water depth stored in the planned wetting layer after irrigation, mm, Z_q is the maximum storable irrigation water depth in the planned wetting layer, mm.

Studies indicate that regional differences and irrigation methods significantly affect field water use efficiency (Fig. 4). Drip irrigation typically demonstrates the highest efficiency, with a global average ranging from 0.88 to 0.94, particularly peaking at 0.94 in the Middle East. Sprinkler irrigation follows, averaging between 0.64 and 0.86, while surface irrigation has the lowest efficiency, averaging 0.42–0.55, with the lowest recorded at 0.47 in South Asia (Jägermeyr et al., 2015).

Field irrigation efficiency is influenced by several factors, including the type of irrigation system—drip, sprinkler, or flood (Levidow et al., 2014)—the properties of the soil, such as permeability and texture (Pereira and Marques, 2017), climatic conditions like precipitation and temperature patterns (Kebede et al., 2014), management practices including irrigation scheduling and real-time monitoring, and water quality factors such as sediment, salinity and nutrient content. Technological advancements, such as water-saving irrigation equipment design for components like drippers, sprinklers, micro-sprinklers, pipeline layouts, filtration systems, and pressure regulators, combined with smart control systems, can further enhance efficiency by improving uniformity and reducing water loss (Liang et al., 2021). Given the complexity of these factors, developing better equipment and strategies to improve irrigation efficiency remains a critical challenge in sustainable agricultural water management.

Measuring field irrigation efficiency has evolved through various methods. Traditional approaches use advance-infiltration models with basic field data to calculate border-irrigation efficiencies (Canone et al., 2015). Deterministic and stochastic models assess irrigation water delivery cost efficiency, aiding resource optimization (Pereira and Marques, 2017). Research shows that sprinkler systems enhance water use efficiency compared to conventional methods. Drip irrigation, conversely, enables precise water delivery directly to the narrow root zone, offering the potential for optimized water management with no yield penalty (Karagiannis et al., 2003). However, a persistent and critical challenge is the difficulty of measuring irrigation water volumes in real-time at the field scale, which is a key contributor to irrigation overuse. This often necessitates reliance on indirect calculation methods (Howell, 2003). Economic efficiency metrics link irrigation outcomes to water allocation (Bosch et al., 1987), and advanced econometric models analyze effectiveness at the farm level (Chatzimichael et al., 2020), particularly regarding technology adoption. The absence of standardized performance indices requires context-specific adjustments, and accuracy assessments are essential (Burt et al., 1997), with error analyses underscoring the need for precision (Clemmens and Burt, 1997). Future studies should develop low-cost methods, particularly using remote sensing, to precisely estimate water use across all irrigation types. This is a critical step toward accurately evaluating and optimizing WUE at the meso-scale (Jensen, 2007).

2.3. Macroscale perspective on water delivery efficiency

To evaluate irrigation efficiency, three key coefficients are used: the canal system water use coefficient, the field water use coefficient, and the irrigation water use coefficient. The canal system coefficient measures the efficiency of water conveyance, calculated as the ratio of net water entering the field to the water drawn from the source. The field water use coefficient assesses irrigation efficiency in the field. The irrigation water use coefficient reflects overall water delivery performance at the irrigation district scale, defined as the ratio of irrigation water stored in the crop root zone to the water drawn from the source (Fan

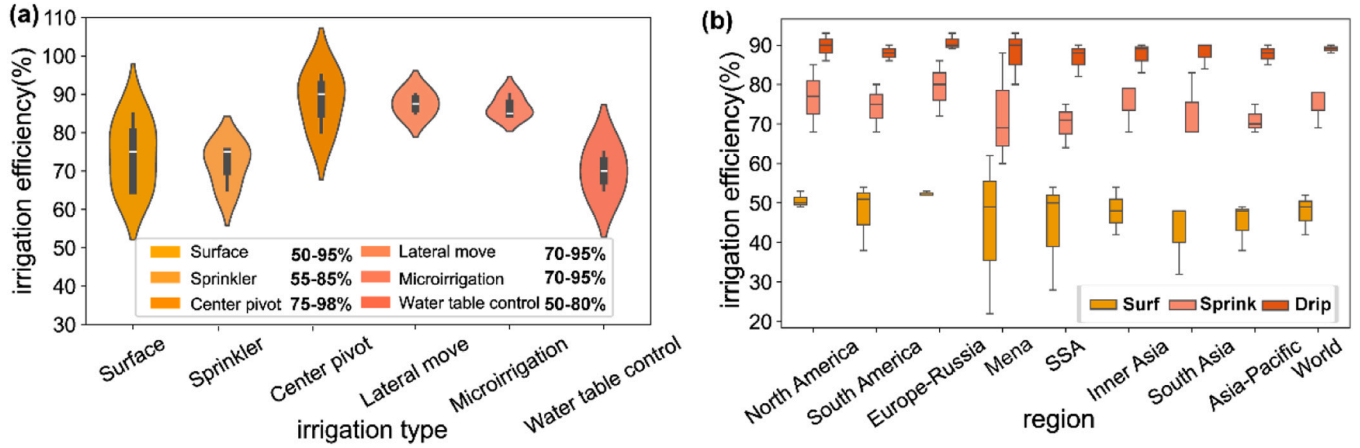


Fig. 4. | Ranges of irrigation efficiency for various irrigation techniques and geographical locations. The data is compiled from multiple references. See the appendix for specific data. (a) Irrigation efficiency across various irrigation techniques. (b) Irrigation efficiency across different geographical areas.

et al., 2024). This coefficient indicates irrigation water conveyance efficiency at the irrigation district scale, and is primarily used to evaluate engineering performance and management effectiveness rather than crop water use.

The canal water use coefficient is the ratio of the net flow, representing the outflow, to the gross flow, corresponding to the inflow, at a specific canal section (Nam et al., 2016). This coefficient reflects the water loss during conveyance in a specific canal segment, and the formula is expressed as follows :

$$\eta_{canal} = \frac{Q_{net}}{Q_{gross}} = \frac{Q_{downstream}}{Q_{upstream}} \tag{8}$$

Where η_{canal} is the canal water use coefficient, Q_{net} and Q_{gross}refer to downstream net flow, while $Q_{downstream}$ and $Q_{upstream}$refer to upstream gross flow.

The canal system water use coefficient reflects the water conveyance losses from the main canal to the field irrigation channels, representing the overall water use efficiency of the entire canal system. Its value is equal to the product of the canal water use coefficients of all the channels operating simultaneously at different levels (Nair et al., 2013). The formula is as follows:

$$\eta_{canal\ system} = \eta_{main\ canal} \cdot \eta_{branch\ canal} \cdot \eta_{distribution\ canal} \cdot \eta_{field\ canal} \tag{9}$$

Where $\eta_{canal\ system}$ is the canal system water use coefficient, $\eta_{main\ canal}$, $\eta_{branch\ canal}$, $\eta_{distribution\ canal}$, $\eta_{field\ canal}$ are the water use coefficient of the main canal, the branch canal, the distribution canal and the field canal.

The irrigation water use coefficient for the entire irrigation district ($\eta_{\text{irrigation water}}$) is the ratio of the net water required in the field ($Q_{field\ net}$) to the water introduced at the head of the canal ($Q_{headwater}$), or it is equal to the product of the canal system water use coefficient ($\eta_{canal\ system\ water}$) and the field water use coefficient ($\eta_{field\ water}$) (Nair et al., 2013). The formula for the irrigation water use coefficient or conveyance efficiency is as follows:

$$\eta_{irrigation\ water} = \frac{Q_{field\ net}}{Q_{headwater}} = \eta_{canal\ system\ water} \cdot \eta_{field\ water} \tag{10}$$

Research shows that the efficiency of water conveyance in irrigation districts is significantly influenced by the type of seepage control used (Fig. 5). The conveyance efficiency of traditional earthen canals with stone masonry seepage control ranges from 0.7 to 0.83, while concrete lining improves efficiency to between 0.84 and 0.89 (Wu and Zhou, 2011). Regionally, Israel leads in effective irrigation water conveyance efficiency at 0.87, followed by Australia and Russia at approximately 0.8. Other countries, including the United States, France, Spain, and Algeria, report efficiencies above 0.62 (Geng et al., 2022), while the irrigation water use coefficient of China is approximately 0.6 (Yan et al., 2024).

Multiple factors affect water conveyance efficiency, including canal design characteristics—such as optimal slope and cross-sectional geometry (T. Li et al., 2018)—and infrastructure maintenance practices (Orojloo et al., 2018), which involve sediment removal and crack repair to minimize leakage. Effective operational management (Molden and Gates, 1990), including real-time flow regulation and demand-based water allocation, also plays a crucial role. Environmental conditions, such as evaporation rates and climate variability, further influence efficiency, alongside technological interventions like automated control systems aimed at reducing losses (Levidow et al., 2014).

Research on irrigation water delivery efficiency combines traditional hydraulic metrics with modern precision techniques. A prevalent method assesses efficiency by multiplying the suitability of the water delivery schedule with the ratio of actual to planned irrigation areas (Clemmens and Bos, 1990). This approach directly reflects irrigation system performance, particularly in aligning water allocation with crop needs and management effectiveness (Rao, 1993). Infrastructure upgrades, such as automated canal controls, have demonstrated the potential to boost efficiency from 65% to over 90% in enhanced systems (Koech and Langat, 2018). Remote sensing facilitates spatial analysis of water distribution across irrigated areas, promoting equitable resource allocation (Bastiaanssen and Bos, 1999). Recent studies recommend integrating hydraulic efficiency metrics with water accounting frameworks at collective irrigation scales (Ferreira et al., 2023). Current guidelines emphasize the need to merge conventional indicators, like conveyance efficiency, with precision technologies to tackle both technical and operational efficiency challenges (Bwambale et al., 2022).

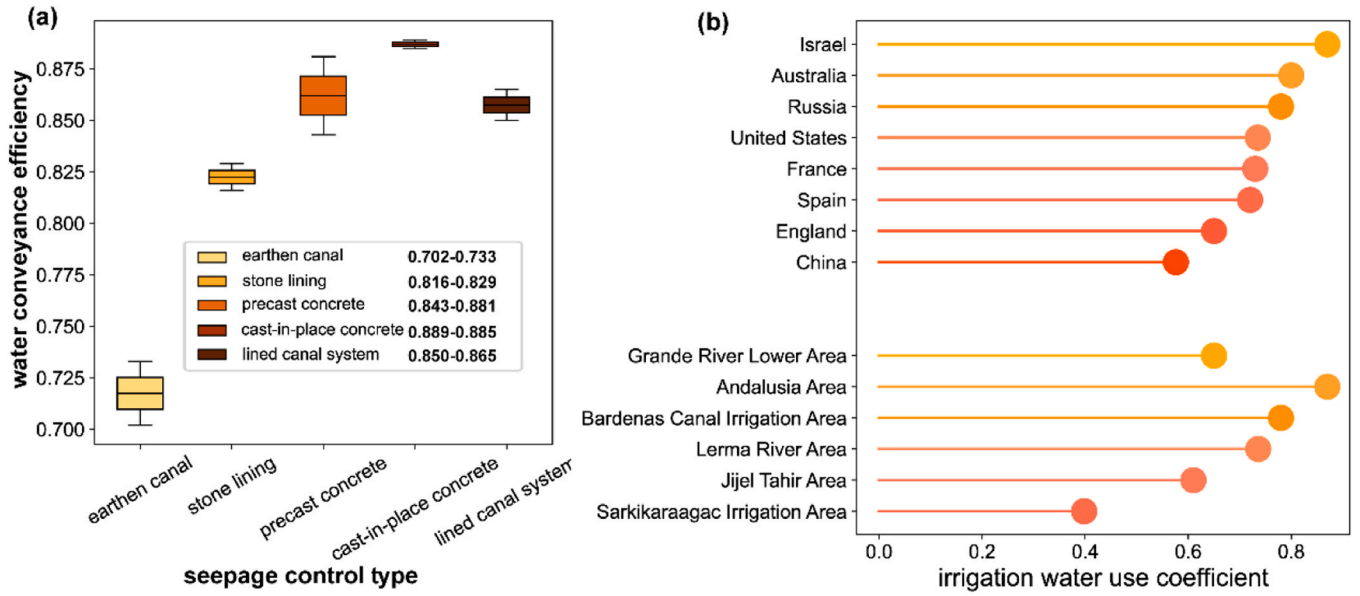


Fig. 5. | Range of water conveyance efficiency and irrigation water use cofficient in irrigation districts. The data is compiled from multiple references. See the appendix for specific data. (a) Water conveyance efficiency under different seepage control types. (b) Irrigation water use cofficient in different regions.

3. Current constraints and trends regarding agricultural water use efficiency and its targeted regulation across scales

3.1. Agricultural water use efficiency in crops

At the microscale, biological water saving focuses on the crop itself (Zhang et al., 2007). This concept involves harnessing and enhancing the physiological and genetic potential of organisms to achieve greater agricultural output with the same water supply (Shao et al., 2007). Accordingly, crop water use efficiency is governed by multi-level biological processes operating from cellular to whole-plant scales under water-limited conditions.

3.1.1. Physiological constraints in crop water use

Crops exhibit growth redundancy as an adaptive strategy for survival amidst environmental fluctuations. While this enhances stability, it can compromise resource-use efficiency under intensive cultivation (Zhang et al., 2022). Precision management techniques aim to strategically reduce this redundancy, optimizing energy allocation for improved yield and quality (Han et al., 2005). Controlled water stress can activate compensatory physiological mechanisms, such as drought-responsive root signaling mediated by abscisic acid (ABA), which optimizes stomatal regulation to minimize transpiration while sustaining photosynthetic activity (Li et al., 2019). Regulated deficit irrigation has been shown to enhance water conservation and yield through post-stress recovery mechanisms (Chen et al., 2023; Dong et al., 2019). However, the molecular basis underlying drought-induced adaptation and compensation remains incompletely resolved. A deeper understanding of their molecular mechanisms is essential to enable effective regulation of plant water-use efficiency (X. Liu et al., 2022; P. Zhang et al., 2023).

Crop water use efficiency is governed by integrated physiological processes, including stomatal dynamics (Kashtoh and Baek, 2021), root architecture optimization, and cuticle development. Stomatal regulation balances transpiration and photosynthesis (WANG et al., 2019), and genetic interventions targeting regulators such as AtMYB60 can reduce stomatal aperture and water loss under drought while maintaining carbon assimilation (Cominelli et al., 2005). Root traits including depth, branching, and root-to-shoot ratio critically influence water acquisition and are regulated by genes such as DRO1 and OsNAC9/TaMOR through gravitropic and hormonal pathways (Li et al., 2016; Redillas et al., 2012; Uga et al., 2013). Concurrently, the cuticle, composed of waxes and cutin, limits non-stomatal water loss (Zhu and Xiong, 2013), with ABA signaling enhancing wax biosynthesis and cuticular integrity via transcription factors such as WIN1/SHN1 (Yang et al., 2011).

At the cellular level, drought sensing involves osmotic pressure detection by calcium channel proteins such as OSCA1 (Parsegian et al., 1995; Yuan et al., 2014), activating ABA-dependent and independent signaling pathways that regulate osmoprotectant synthesis and ion homeostasis to maintain turgor pressure (Seki et al., 2007; Zhu, 2001). Stress-responsive proteins, including aquaporins and LEA proteins, further support cellular water balance (Shinozaki and Yamaguchi-Shinozaki, 2007), while antioxidant systems mitigate drought-induced reactive oxygen species (Kliebenstein et al., 1998; Tekle and Alemu, 2016). Hormonal coordination integrates these responses, with ABA governing stomatal closure and osmotic adjustment (Daszkowska-Golec, 2016), and auxins, brassinosteroids, and cytokinins modulating root plasticity, antioxidant capacity, and stress-responsive gene expression (Kilemo, 2022; Ullah et al., 2018). The complex hormonal cross-talk underlying optimal WUE across variable environments is still not fully understood.

3.1.2. Multi-omics-driven crop water use decoding

Crop phenotyping is essential for capturing genotype–environment interactions and unlocking the genetic basis of WUE (Marsh et al., 2021). High-throughput phenotyping platforms integrating multispectral imaging, lidar, and micro-CT enable characterization from cellular to canopy scales (D. Li et al., 2021; Zhao et al., 2019), supporting WUE prediction using thermal imaging, VNIR spectroscopy (Wang et al., 2016), and root water uptake quantification (Yan et al., 2022). Omics approaches, including transcriptomics, proteomics, and metabolomics, have identified regulatory networks (Baldoni et al., 2021; Deng et al., 2026; Karami et al., 2025), aquaporins and water channel proteins (Goche et al., 2020; Kausar et al., 2022), and metabolic reprogramming associated with enhanced WUE (Anzano et al., 2022; Yang et al., 2025). These integrative datasets reveal complex genotype-by-environment (G×E) interactions shaping WUE (Xiao et al., 2024) (Fig. 6).

Emerging single-cell and spatial omics technologies provide finer resolution of drought-responsive regulation (Yin et al., 2023; Yuan et al., 2018), while epigenetic mechanisms such as DNA methylation and non-coding RNAs contribute to dynamic and transgenerational regulation of drought-adaptive traits (Islam et al., 2022; Jha et al., 2020; Q. Wang et al., 2021). Genomic tools, including whole-genome resequencing, GWAS, QTL mapping, and long-read sequencing, have accelerated the identification of drought-resistant genes and structural variants (Ansari et al., 2024; Xu et al., 2022). "Super pan-genome" strategies further expand the genetic pool by capturing stress-resilient alleles from wild relatives (Brink, 2021; Marsh et al., 2021). These approaches hold great promise for expanding the genetic base for WUE, because dissecting these molecular mechanisms is critical for translating physiological regulation of photosynthesis and transpiration into breeding targets; however, integrating genes from wild relatives into cultivated crops without compromising agronomic performance remains technically challenging.

The integration of multi-omics data with artificial intelligence and deep learning has enhanced phenotyping, genomic selection, and regulatory network inference for WUE improvement (Xie and Zhang et al., 2023; Mastrangelo et al., 2011; Lee et al., 2018; Zhang et al., 2023; Dubois et al., 2017; López et al., 2022; Rohwer, 2012; Wu et al., 2015). These approaches enable system-level interpretation of drought adaptation, including epigenetic regulation dynamics (López et al., 2022), and provide new analytical frameworks for water-efficient crop breeding (Aerts et al., 2024; Sharma et al., 2023).

3.1.3. Regulation measures of crop water use efficiency

Crop WUE can be regulated internally through biotechnological approaches, including genetic engineering and synthetic biology. Targeted manipulation of drought-responsive transcription factors, osmotic protection genes, and regulators of root architecture and stomatal movement has improved drought resilience and WUE (X. Jia et al., 2025; Yang et al., 2010; Ruggiero et al., 2017). Synthetic biology further enables refined regulation through engineered carbon-concentrating mechanisms, synthetic promoters, and artificial gene circuits controlling stomatal behavior (Borland et al., 2014; Lim et al., 2019; Yang et al., 2015; Amin et al., 2019; Karavolias et al., 2024; Andres et al., 2019; Schmal et al., 2013). Advanced techniques such as optogenetics provide precise temporal control of stomatal dynamics and WUE (Papanatsiou et al., 2019; Nguyen et al., 2023).

Externally, crop WUE can be enhanced through chemical, nanotechnological, and emerging biophysical interventions. Nanoparticles and nano-carriers improve photosynthetic efficiency and drought resistance (Kang et al., 2025; Raza et al., 2023), exemplified by ZnO-NPs in eggplant (Semida et al., 2021) and SiO_2-NPs in wheat (Akhtar et al., 2021). Carbon nanotubes enhance antioxidant capacity in maize and soybean (Shahriari et al., 2019; Wenli et al., 2020), while nanoparticle-mediated delivery systems facilitate precise CRISPR/Cas9 gene editing (Cheng et al., 2020; Demirer et al., 2020; Qiu et al., 2021; Ahmar et al., 2021). Phytohormones, including gibberellins, auxins, cytokinins, and jasmonates, regulate growth, photosynthesis, and antioxidant activity under drought stress (Javid et al., 2011; Raza et al., 2022; Kumari et al., 2018; Ghaffari et al., 2019; Mousavi et al., 2022). Emerging biophysical approaches, such as magnetic field and electromagnetic treatments, have also been reported to modulate plant

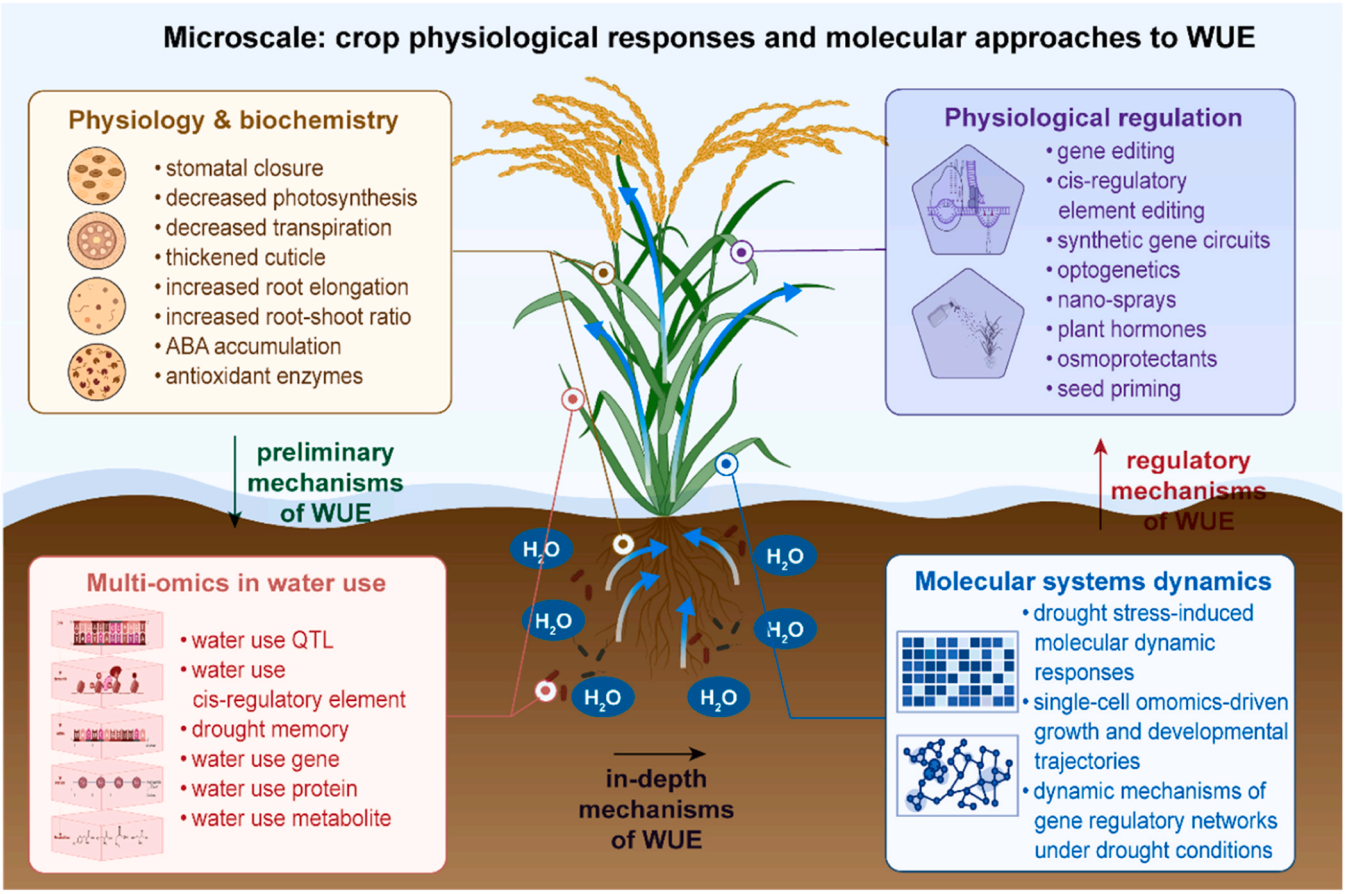


Fig. 6. | Evaluation and regulation of water use efficiency at the microscale. This figure presents a crop-centric perspective, focusing on the understanding and manipulation of microscopic mechanisms governing crop water use.

physiological processes related to water use under controlled conditions (Sharafi, 2025; Zablotskii et al., 2025). Osmoprotectants and seed priming further enhance drought tolerance, WUE, yield, and seed quality (Ashraf and Foolad, 2007; Elkelish et al., 2020; Sallam et al., 2019; Seleiman et al., 2021; Nawaz et al., 2016; Freitas and da Silva, 2024; Khan et al., 2020).

3.2. *Agricultural water use efficiency in fields*

At the mesoscale, research primarily targets farmland, focusing on soil moisture monitoring, water simulation processes, water-efficient irrigation techniques, water saving equipment, and soil improvement strategies. Common agronomic practices, such as mulching, soil structure enhancement, organic fertilizer application, optimized planting density, crop rotation, and integrated water-fertilizer management, are widely recognized for reducing non-productive water losses and improving water use efficiency. Advanced irrigation technologies, including alternate furrow irrigation and micro-irrigation systems, are prioritized alongside improved surface irrigation scheduling to maximize water use efficiency. Despite their demonstrated effectiveness, the performance of these approaches is often constrained by field heterogeneity and management complexity.

3.2.1. *Technical and management constraints in field water use*

Achieving effective water savings at the field scale is closely linked to accurate monitoring and simulation of soil water dynamics. While real-time soil moisture sensors and hydrological models support irrigation scheduling, their application is frequently limited by cost, variable sensor reliability, and site-specific calibration requirements associated with heterogeneous soil properties and root distributions (Abioye et al., 2020; Viani et al., 2017). Mismatch between simulated and actual soil water movement can result in deep percolation losses or unintended crop water stress, reducing irrigation efficiency.

Optimizing irrigation system design remains essential, with drip and sprinkler systems selected according to topography, soil texture, and crop canopy characteristics (Zinkernagel et al., 2020). However, uneven emitter spacing, pressure imbalance, and suboptimal installation commonly reduce uniformity and offset potential water savings. Although sensor-assisted control and automated pressure regulation allow dynamic adjustment, their effectiveness depends on accurate local calibration, including emitter placement relative to root zones. Hybrid irrigation systems combining drip and sprinkler methods provide flexibility under variable rainfall and crop development, while sprinkler irrigation may additionally mitigate heat stress through canopy cooling.

Irrigation timing and volume represent persistent management challenges. Daytime irrigation under high temperatures increases evaporative losses, whereas nighttime or early-morning irrigation may conflict with labor or energy availability (Zinkernagel et al., 2020). Automated weather-based irrigation controllers improve scheduling accuracy, but adoption remains uneven due to technical skill requirements and capital costs (Bwambale et al., 2022). Uniform irrigation prescriptions often fail to capture within-field variability, leaving considerable efficiency gains unrealized (Evans and King, 2010).

3.2.2. *Field water monitoring, simulation and water saving irrigation*

The integration of sensor networks and remote sensing has significantly advanced precision soil moisture monitoring and irrigation management (Bwambale et al., 2023; J. Wang et al., 2024). Sensor systems provide real-time, multi-depth soil moisture data, enabling adaptive irrigation strategies aligned with crop water demand (Jin et al.,

2025). Satellite and UAV-based remote sensing further support high-resolution monitoring of evapotranspiration and crop water stress (Antoniuk et al., 2021; Kilemo, 2022), informing site-specific irrigation during sensitive growth stages (Deng et al., 2025, 2024; Hu et al., 2024; Kilemo, 2022; Yang et al., 2019). Deep learning methods enhance analysis of water-salt dynamics and drought response (Ding et al., 2021; Morianou et al., 2025; Yuan et al., 2025). Modern simulation tools combine physics-based models like HYDRUS and SWAP with data-driven and hybrid approaches to evaluate soil–water transport and irrigation performance (Yanling Wang et al., 2023b; L. He et al., 2024; Yanling Wang et al., 2023a). Crop water-use models such as AquaCrop and SWAT link climate, water, and yield responses, supporting irrigation design and water allocation decisions (Ceglar et al., 2019; Vanuytrecht et al., 2014). Together, these tools enable feedback-driven irrigation management, although large-scale, real-time implementation remains technically demanding (Wang et al., 2025) (Fig. 7).

Field irrigation technologies include surface, sprinkler, and localized systems (Deng et al., 2017). Sprinkler irrigation offers high uniformity (Chauhdary et al., 2023), while laser-leveled surface irrigation improves distribution efficiency (Jat et al., 2006). Micro-irrigation systems enhance precision for high-value crops (Evans and Sadler, 2008; Smith and Baillie, 2009), though initial investment and clogging risks limit adoption in some regions. Subsurface drip irrigation reduces evaporation and improves productivity in arid areas (Qi Xu et al., 2024), but requires careful installation and maintenance. Physiology-based irrigation strategies, including regulated deficit and alternate root-zone irrigation, improve water productivity by exploiting crop compensatory responses, though their effectiveness depends on accurate monitoring and management (Cheng et al., 2021; Crookston et al., 2025).

Water-fertilizer coupling further enhances efficiency at both rhizosphere and field scales. Optimized irrigation improves soil microbial activity and nutrient cycling under drought stress (Bandopadhyay et al., 2024; Xu et al., 2021, 2018; P. Chen et al., 2024; F. Jia et al., 2025). Integrated practices such as intercropping, crop rotation, and spatio-temporal irrigation-fertilizer coordination improve soil water retention and overall water use efficiency (Liou and Kar, 2014; Lu et al., 2000; McShane et al., 2017; Yuan and Shen, 2013; Zhang et al., 2001; Mao et al., 2015; Tan et al., 2021). Multi-objective optimization strategies targeting water conservation, yield stability, and emission reduction show promise, although trade-offs remain (J. Li et al., 2018).

3.2.3. *Regulation measures of field irrigation efficiency*

Precision irrigation aims to align water delivery with crop demand through optimized timing and application rates (Feng, 2011; Shuwen

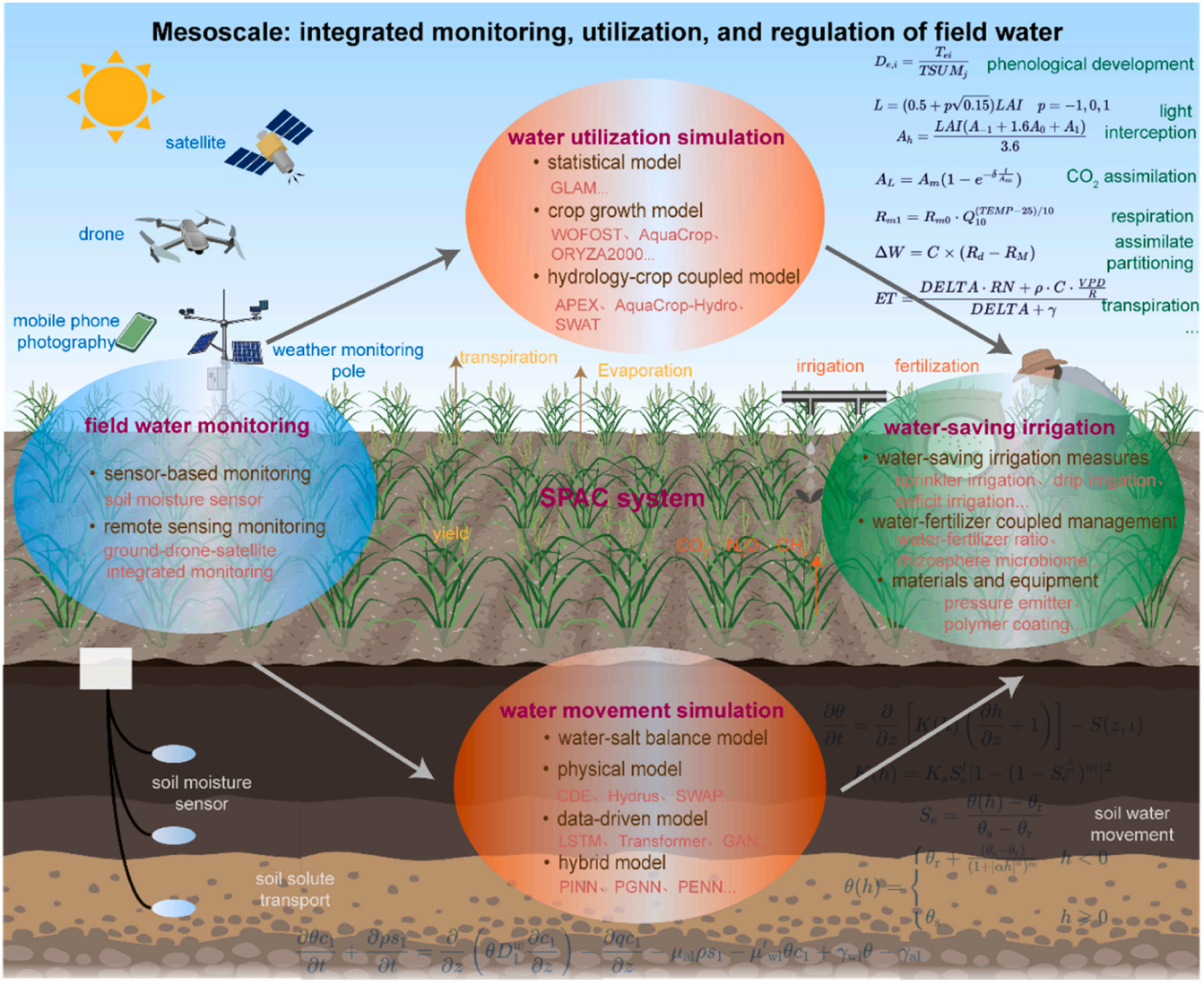


Fig. 7. | Evaluation and regulation of water use efficiency at the mesoscale. From the perspective of agricultural fields, this figure focuses on monitoring moisture and water use, simulating and understanding the mechanisms of moisture and water utilization, and achieving better regulation of water-saving irrigation practices.

and Changli, 2015; Singh et al., 2019). Zonal irrigation management enables site-specific scheduling based on soil, topography, and crop growth stage (Choudhary et al., 2019), but requires high-resolution data and advanced analytics. Equipment innovations—including pressure-compensating emitters, anti-clogging drippers, subsurface systems, and adjustable sprinkler layouts—improve uniformity and reduce runoff and deep percolation losses.

Modern irrigation infrastructure increasingly incorporates durable materials and optimized designs. Pressure-compensating emitters enhance system reliability (Shamshery and Winter, 2018), while polymer-coated micro-sprinklers improve performance under saline conditions (J. Liu et al., 2022). Drip irrigation combined with plastic mulching effectively reduces evaporation (Garcia-Caparros et al., 2017), though environmental concerns related to plastic use remain. Fertigation enables synchronized water and nutrient delivery, supported by soil moisture and crop nutrient sensing, while solar-powered systems offer energy-efficient alternatives. However, system complexity and data integration remain limiting factors.

Soil-focused agronomic measures complement irrigation technologies. Organic mulching, superabsorbent polymers, and bio-based materials improve soil water retention (Choudhary et al., 2019; Patra et al., 2022; Zheng et al., 2023; Demo and Asefa Bogale, 2024; Haider et al., 2020). Microbial amendments such as mycorrhizal fungi further enhance soil structure and water availability (Wu et al., 2022). Overall, while integrated field-scale technologies substantially improve agricultural water use efficiency, their effectiveness remains shaped by economic feasibility, management capacity, and site-specific conditions.

3.3. Agricultural water use efficiency in irrigation district

At the macroscale, research focuses on optimizing water resource management in irrigation districts. This encompasses monitoring regional agricultural and water conditions, precise resource allocation, irrigation forecasting, dynamic decision-making, and the development of efficient water conveyance infrastructure. The rapid advancement of intelligent technologies has made digital water resource scheduling, precision irrigation, and inter-basin water management essential strategies for improving water efficiency and ensuring agricultural sustainability.

3.3.1. Systemic and engineering constraints in irrigation district water use

Despite progress in digital water resource management, dynamic regional water allocation remains constrained by systemic and engineering limitations. A primary challenge lies in integrating heterogeneous crop, atmosphere, soil, and aquifer data with limited spatiotemporal resolution, which restricts the reliability of multi-source water scheduling models. For example, synchronizing crop planting structure optimization with groundwater recharge management remains difficult (Zhang and Guo, 2016), highlighting persistent data and coordination gaps.

Real-time dynamic allocation models at the regional scale are computationally intensive, as large irrigation networks with numerous delivery nodes and diverse cropping patterns generate high-dimensional, combinatorial optimization problems (Yafei Wang et al., 2023). As a result, highly precise solutions are often impractical for operational deployment. Even when multi-stage coupling approaches are applied to inter-basin transfers, reconciling delayed reservoir responses with immediate farm-level demands remains challenging. The non-convexity of allocation problems and nonlinear crop yield responses frequently necessitate simplified approximations, reducing solution fidelity (Mai et al., 2024).

Beyond technical complexity, regional water allocation must balance economic returns, ecological requirements, and social equity (Kaarthikeyan and Suresh, 2019). This multi-objective nature introduces governance and policy constraints that cannot be resolved through engineering solutions alone. Meanwhile, aging conveyance infrastructure—such as unlined earthen canals—continues to cause substantial seepage and evaporation losses. Although lining and pressurized systems offer clear efficiency gains, high costs, difficult terrain, and regional funding disparities limit large-scale upgrades, as illustrated by the Hetao Irrigation District (Miao et al., 2015). Environmental constraints, including wind effects on sprinklers, soil salinization, and groundwater contamination from over-extraction, further restrict the performance of high-efficiency irrigation technologies (Levidow et al., 2014).

3.3.2. Reginal monitoring, forecasting and water allocation decision-making

Advances in multi-source remote sensing and information technologies have substantially improved regional agricultural monitoring. The integration of satellite and UAV imagery, including thermal infrared data, enables effective detection of soil moisture status and early crop water stress across scales (MA et al., 2019; Wang et al., 2013; Yuan et al., 2019). Vegetation indices such as NDVI further support real-time assessment of crop water use efficiency (El-Shirbeny et al., 2021; Lin et al., 2017). Near-ground sensing and sky–ground integrated networks enhance precision irrigation zoning by linking multi-scale observations with crop growth models (An et al., 2025; Liu et al., 2022). However, data heterogeneity and limited model generalization remain persistent operational challenges.

Hydrological modeling for irrigation districts increasingly combines mechanistic and data-driven approaches. Physically based models such as SWAT simulate runoff and pollutant transport effectively (Hexiang et al., 2025; Ikenberry et al., 2017), while grid-based models improve performance in complex terrain (Hu et al., 2006). Data-driven methods, including Bayesian frameworks and deep learning, enhance long-term runoff forecasting and extend modeling capacity to data-scarce regions (Tabas and Samadi, 2022; Ding et al., 2020; Wei and Xu, 2025). Hybrid models that couple physical processes with machine learning balance interpretability and adaptability, as demonstrated by deep learning–HEC-HMS integrations (Debbarma et al., 2025; Karpatne et al., 2024). Remote sensing assimilation further improves soil moisture prediction and irrigation demand estimation (Corbari et al., 2019), while nested real-time forecasting systems support dynamic irrigation adjustments (Ceppi et al., 2014). Despite these advances, operational interoperability across scales remains limited.

Decision-making in irrigation districts increasingly relies on integrated optimization frameworks. Linear, bi-level, and mixed-integer nonlinear programming support multi-sectoral water allocation (S. Zhang et al., 2023; Ren et al., 2023; Abdulbaki et al., 2017), while intelligent algorithms enable dynamic canal operation (R. Li et al., 2021; Mai et al., 2024). Reinforcement learning offers real-time adaptive control for balancing yield and water stress (Kelly et al., 2024). Although promising, these approaches remain computationally demanding for large-scale systems, and their effectiveness depends on high-quality, spatially explicit data (GU et al., 2003; Agyeman et al., 2024). Collectively, these developments shift irrigation management toward adaptive, system-level optimization (Fig. 8).

3.3.3. Regulation measures of water delivery efficiency

Advanced observation, simulation, and data assimilation technologies are reshaping regional water regulation by enabling near-real-time modeling of soil moisture, crop demand, and aquifer dynamics (Al Mamun et al., 2025; Goap et al., 2018; Mengistu et al., 2022; X. Wang et al., 2021). Machine learning-enhanced hydrological models facilitate scenario-based allocation planning that balances agricultural production with ecological flow requirements (Chatrabhuj et al., 2024; Li et al., 2023; Shaikh and Birajdar, 2024). Sensor-driven irrigation systems dynamically adjust scheduling based on evapotranspiration and soil moisture, improving efficiency while protecting environmental water reserves (Huan and Meng, 2021; Mengistu et al., 2022). However, high costs, maintenance demands, and connectivity requirements

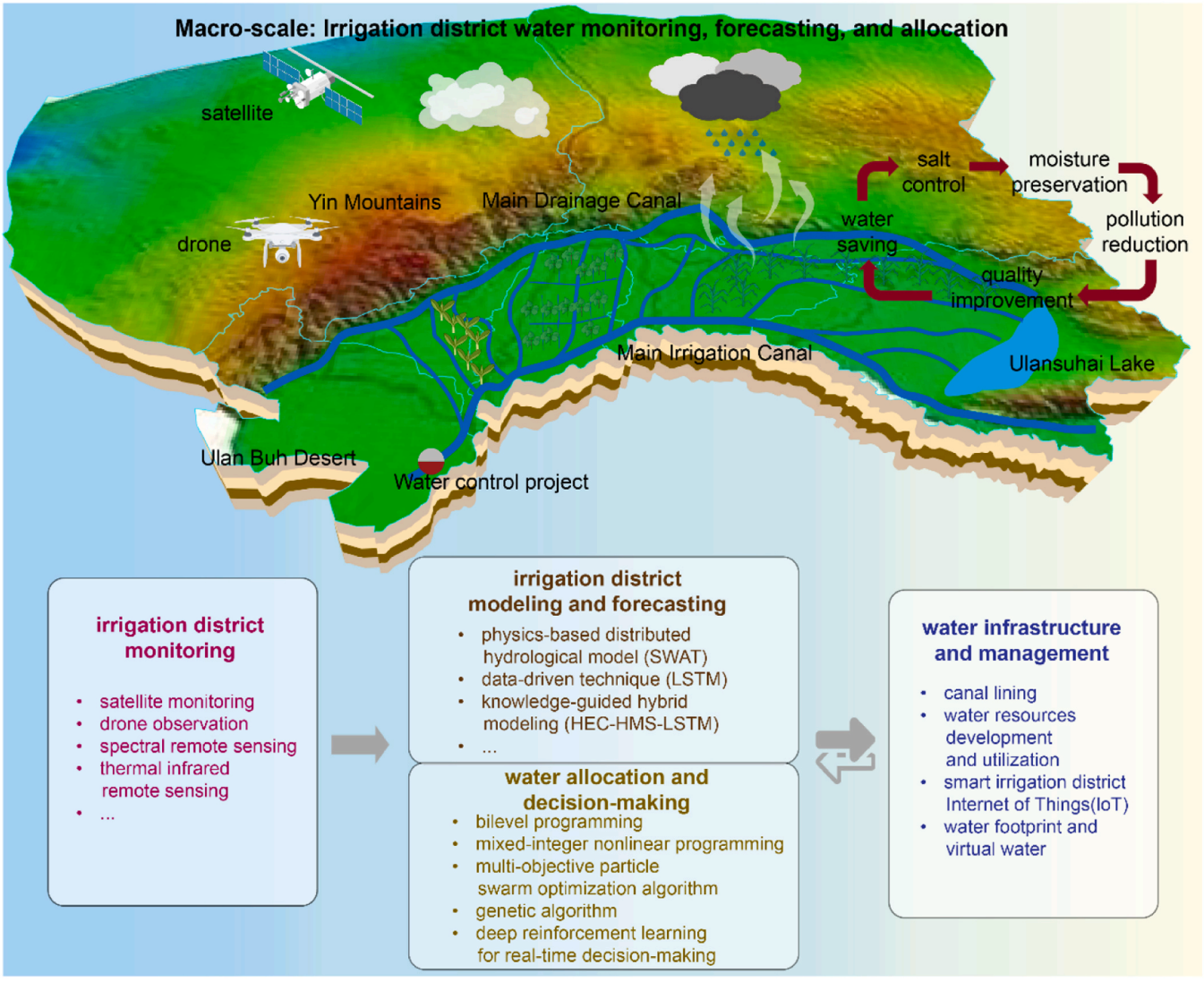


Fig. 8. | Evaluation and regulation of water use efficiency at the macroscale. The Hetao Irrigation District, dating back to the over 2200 years, has developed a seven-tier water network: 1 main trunk canal feeds 13 primary canals, 48 sub-primary canals, 339 branch canals, and 85,522 field-level channels (dou/nong/mao), delivering water to approximately 12 million mu (1.2 million hectares) of farmland. Drainage occurs via 17,322 field-level ditches, 346 branch ditches, 64 sub-primary ditches, and 12 primary ditches, converging into the main drainage trunk and Ulatai Sea before returning to the Yellow River. Managing 4 billion m^3 of annual water quotas across this arid-region system poses significant challenges while ensuring ecological security. The irrigation efficiency of the Hetao Irrigation District is composed of two parts: conveyance efficiency and application efficiency. For example, water resource utilization analysis in wheat production indicates a conveyance efficiency of 52.08% and an application efficiency of 76.09%. Conveyance losses constitute the majority of water loss during wheat production. Therefore, reducing this portion of water loss to improve WUE is an urgent problem to be solved.

constrain large-scale deployment, particularly in resource-limited regions.

Material and system innovations further enhance water delivery efficiency. Smart polymer composites improve canal seepage control and enable adaptive permeability (Sun et al., 2014; Wu-quan et al., 2024), while BIM-GIS platforms integrate infrastructure data with hydrological models to create digital twins that simulate water-energy-carbon interactions (Li and Ma, 2025). Despite their analytical power, these systems face computational and standardization constraints. Automated drought-response mechanisms—such as ecological flow prioritization and tiered water pricing—have demonstrated effectiveness in arid regions (Abdulla et al., 2021; Dotaniya et al., 2023; Oster, 1994; Puppala et al., 2023), though social acceptance and institutional feasibility remain critical barriers, especially in rural areas where farmer decision-making capacity strongly influences outcomes (Guo et al., 2025; Yuan et al., 2021).

Virtual water trade analyses provide complementary system-level regulation by quantifying how crop structure adjustment and water recycling redistribute embedded water resources (Bazrafshan et al., 2020; Novoa et al., 2023; Tian et al., 2018). At this broader scale, agricultural water use efficiency is increasingly evaluated using integrative frameworks, including the water-energy-food nexus (Endo et al., 2017), blue and green water accounting (Kilemo, 2022), water footprint analysis (Hoekstra, 2017), and life cycle assessment (LCA) (Berger and Finkbeiner, 2010). These approaches contextualize WUE improvements within cross-sectoral interactions and spatial water redistribution (Falkenmark, 2020). Although not direct efficiency metrics, they define system-level constraints under which WUE gains can translate into actual water saving.

4. Challenges and opportunities in breaking limits of agricultural water use efficiency

Enhancing water use efficiency in agriculture faces multifaceted

challenges across biological, engineering and systemic dimensions. Genetically, incomplete understanding of regulatory networks and gene-environment interactions limits the development of drought-resistant and water-efficient traits (Sharma et al., 2023). Emerging technologies like CRISPR, nanomaterials, and optogenetics show promise in labs but struggle with biosafety, yield penalties, scalability, and cost in field applications (Ning et al., 2025). While emerging water dynamics simulation model, irrigation technologies, soil amendments, and integrated water-fertilizer approaches demonstrate progress in farmland water management (Taotao et al., 2024), key challenges persist. These barriers include methodological gaps in scaling results across spatial scales, compounded by practical issues like infrastructure deficits and soil variability, as well as socioeconomic hurdles such as farmer resistance to initial investments. Balancing water saving with yield stability remains constrained by technical accessibility, spatial heterogeneity, and socioeconomic barriers to sustainable practice implementation. Regionally, a critical challenge lies not only in optimizing water allocation, but in determining how complex real-world water processes can be represented in a scientifically consistent yet operationally feasible manner. Dynamic governance is essential to balance ecological constraints like groundwater recharge and soil health with agricultural needs, as unsustainable extraction depletes aquifers and over-conservation disrupts ecosystems. Adaptive frameworks must harmonize agricultural, ecological, and climatic priorities to optimize regional water resource allocation through predictive modeling, precision monitoring, and responsive policies, ensuring water, food, and ecosystem sustainability. Future meta-analyses could quantify the context-specific effects of key WUE-improving strategies, including deficit irrigation, water–fertilizer coupling, crop improvement, and sensor- or AI-assisted irrigation (Baldoni et al., 2021; Cheng et al., 2021; Mutema and Dhavu, 2022).

4.1. Breaking the biological limits of water use efficiency

The biological limit of water use efficiency is the physiological water-saving capacity ceiling formed by crops through long-term evolution, constrained by genetic expression, cellular metabolism, and plant structure (T. Wang et al., 2024). For example, the circadian rhythm of leaf stomatal opening and closing is strictly regulated by genetic networks (Sun et al., 2022), and the maximum daily water uptake of roots is physically constrained by the xylem vessel diameter and the risk of embolism (Kramer, 1932). This limit is essentially the "survival baseline" for plants to adapt to their environment.

Future water-saving agriculture urgently needs to break through the biological limits of crop water use efficiency by increasing biomass and yield per unit water consumed, rather than by expanding total water use. Biological advancements in improving crop water use efficiency are likely to focus on integrating genetic regulation, synthetic biology, and nanomaterial-based bioagents. Synthetic biology approaches, such as designing drought-responsive cis-regulatory elements like ABA-inducible promoters and artificial gene circuits, could enable programmable control of stomatal dynamics and root architecture (Kumar et al., 2021). Metabolic engineering strategies, including transplanting C_4-specific carbon fixation modules like PEP carboxylase into C_3 crops, may reduce photorespiration and enhance carbon assimilation per unit of water input (Kumar et al., 2021). Optogenetic tools, like phytochrome-coupled stomatal regulators, could allow light-controlled transpiration to dynamically balance photosynthesis and water conservation (Kumar et al., 2021). Nanotechnology also offers precision solutions. Silica nanoparticles (SiNPs) and lipid-based nanocarriers can deliver CRISPR components or RNA molecules to specific tissues for localized gene editing, minimizing systemic metabolic disruptions (Pratap et al., 2024). Stimuli-responsive nanosensors, such as mesoporous silica loaded with drought-signaling molecules like ABA, may enable real-time monitoring and adaptive closure of stomata under water scarcity (Goswami et al., 2022). Bio-nano hybrid systems, combining plant-derived biopolymers like chitosan with metallic nanoparticles, could form protective leaf coatings to reduce non-stomatal water loss while enhancing stress resilience (Goswami et al., 2022). However, challenges remain in scaling these technologies. These include biosafety concerns such as nanomaterial accumulation in edible tissues and the need for multi-omics modeling to predict long-term effects of gene-nano interactions (Chen et al., 2024; Jing et al., 2024). Moreover, improvements in intrinsic physiological efficiency at the molecular or organ scale do not automatically translate into yield-scaled or field-scale water savings, as these gains may be offset by compensatory growth responses or management constraints.

4.2. Breaking the abiotic limits of water use efficiency

The abiotic limit of water use efficiency is the engineering maximum achieved through technological innovations at the field and irrigation area scales. This includes the maximum irrigation water saving coefficients in the field and pipeline water distribution efficiency in irrigation district. This abiotic limit primarily arises from human-engineered optimizations in water management. It reflects the physical constraints imposed by advanced irrigation technologies and engineered water distribution networks, where human interventions minimize water losses.

Abiotic limits of water use efficiency are likely broken through innovations in water-saving irrigation equipment, integrated water and fertilizer management, soil improvement strategies, and next-generation modeling of water, solute, crop, and ecosystem dynamics. Innovating irrigation hardware and intelligent control software is essential, though their universal application-especially for marginalized groups at regional scales-faces significant challenges. Future developments must address system limitations, including enhanced adaptability to poor-quality water and improved resilience to concurrent abiotic stresses such as heat stress, which can severely limit production even under full irrigation (Kilemo, 2022). Key advancements should focus on optimizing hydraulic components, enhancing material durability, upgrading self-cleaning mechanisms, and integrating precise water-fertilizer systems. Complementary soil management practices-including deep tillage, organic amendment, biochar application, and microbial regulation-further enhance water use efficiency by improving soil structure and root-zone resilience. Importantly, the adoption of these techniques by marginalized communities requires addressing socioeconomic barriers and recognizing the value of non-centralized approaches. In many such contexts, large-scale canalization, centralized control, or high-capital irrigation infrastructure is neither technically feasible nor economically sustainable, imposing a hard constraint on how close the abiotic WUE limit can be approached. Moreover, the traditional concept of water-saving must evolve into a multidimensional framework that integrates efficient water use with yield, quality, pollution control, and carbon sequestration (Ullah et al., 2017), thereby advancing from mere “yield protection” to amplifying sustainable value per unit of water consumed (Kilemo, 2022).

The efficacy of water saving measures is largely contingent upon the precise estimation of agricultural water consumption and the projection of future water requirements at both field and regional levels. The process hinges on digitizing fields and irrigation districts, followed by simulation, and ultimately, model-driven optimization of regional water distribution decisions. However, high-resolution regional monitoring and dynamic modeling face hard constraints related to data sparsity, parameter uncertainty, and the inherent heterogeneity of soils, crops, and management practices, which limit the transferability of field-scale efficiencies to larger spatial scales. Digital twin systems, representing a sophisticated integration of these elements, can offer a real-time, data-driven virtual replica of the irrigation system, thereby significantly improving decision-making accuracy (Kilemo, 2022, 2022). Yet such systems remain approximations rather than fully predictive representations of real-world hydrological complexity.

4.3. Multi-scale perception of water processes

Future advancements in water process sensing will focus on cross-scale observational integration, spanning molecular to regional levels, to enable rapid and precise detection of water-related signals. It is important to acknowledge that direct upscaling of parameters or governing equations from molecular, leaf, or soil-profile scales to regional scales is physically infeasible, as fundamentally different laws apply across scales. At the microscopic scale, emerging technologies such as fluorescence nanoprobe-based imaging and gene expression biosensors could revolutionize our understanding of water dynamics. For instance, nanoprobes functionalized with water-sensitive dyes or quantum dots may track real-time stomatal aperture changes or xylem sap flow in plants under drought stress, while biosensors could link cellular hydraulic conductivity to drought-responsive gene expression (Kim et al., 2024). To gain a holistic understanding, multi-omics observation – encompassing genomics, transcriptomics, proteomics, and metabolomics – is crucial for characterizing the complex molecular networks involved in WUE. These tools provide cellular-resolution insights into water transport mechanisms, critical for engineering water saving crops. Scaling up to field-level observations, hyperspectral and thermal imaging systems enable non-invasive mapping of canopy water status and transpiration fluxes. Hyperspectral cameras detect subtle shifts in leaf water content by analyzing near-infrared absorption bands, while thermal drones map canopy temperature gradients to infer stomatal conductance. Concurrently, LiDAR-driven 3D root architecture mapping, augmented by ground-penetrating radar, resolves soil-root water uptake patterns, allowing farmers to optimize irrigation strategies based on root-soil moisture interactions (D. Li et al., 2021; Zhao et al., 2019). Autonomous field robots equipped with multi-sensor payloads further bridge plant-scale data with field heterogeneity, enabling precision agriculture. At the regional to continental scale, next-generation satellite constellations with enhanced temporal and spatial resolution, such as hyperspectral sensors and synthetic aperture radar, will transform large-scale water monitoring. Hyperspectral satellites can map soil moisture anomalies and crop water stress at high resolution, while synthetic aperture radar systems penetrate cloud cover to detect subsurface water storage changes (Hansana et al., 2023). Cross-scale integration should be understood not as mechanistic unification, but as information coupling, where fine-scale observations inform constraints, thresholds, and parameter ranges within water accounting and balance-based regional frameworks. This approach enables practical decision-making while respecting physical and methodological boundaries.

4.4. System dynamics for water and accompanying substances at different scales

System dynamics, as a computational framework, originated to model complex systems by integrating feedback mechanisms, nonlinear interactions, and time-dependent variables. In water-related studies, it has been widely applied to simulate soil-water dynamics, such as HYDRUS models focusing on soil moisture transport and solute movement (Ding et al., 2021), and broader agro-hydrological systems, such as the soil-plant-atmosphere continuum conceptualized in models like WOFOST. These models emphasize macroscale interactions among physical, biological, and environmental components. However, classical approaches often overlook finer-scale biological and genetic processes, limiting their ability to mechanistically explain phenomena such as crop water-use efficiency or contaminant transformation. It should be emphasized that system dynamics is not intended to replace scale-specific physical or physiological models, nor to directly upscale governing equations across biological, field, and regional scales (Blöschl and Sivapalan, 1995). Instead, its strength lies in coordinating heterogeneous models and processes that are governed by fundamentally different laws at different scales (Troch et al., 2009). Future advancements must bridge these gaps by extending system dynamics to hierarchical scales—from genes to ecosystems-while explicitly acknowledging the limits of mechanistic continuity across scales.

To enhance predictive accuracy, system dynamics should incorporate gene system dynamics and microbial community dynamics. Gene system dynamics involves modeling how individual or clustered gene expressions regulate plant traits such as stomatal conductance and root architecture under water stress. For instance, linking gene networks to phenotypic outputs, such as drought-induced transcription factors modulating aquaporin activity, could refine crop models by embedding genetic adaptability. Concurrently, microbial dynamics must address how rhizosphere microbiomes mediate nutrient cycling, water retention, and contaminant degradation. However, parameterizing such gene–microbe–plant interactions at system scale remains highly uncertain, as empirical data are often sparse, context-dependent, and difficult to generalize (Filippi et al., 2025). This introduces significant challenges in model calibration, validation, and interpretability. Together, gene–microbe–plant system dynamics should therefore be viewed as a conceptual bridge for scenario analysis rather than a fully predictive framework (Saltelli et al., 2020).

Classical system dynamics often simplifies hydraulic infrastructure as static conduits, neglecting the spatiotemporal complexity of real-world channel and pipeline networks (Loucks and Van Beek, 2017). Future research must adopt network dynamics to simulate water and solute transport across interconnected channels, drains, and urban pipelines (Di Nardo et al., 2015). This requires coupling fluid mechanics with topology optimization, addressing challenges like sediment transport in open channels, biofilm-induced corrosion in pipes, and transient contaminant dispersion. Advanced frameworks could integrate machine learning for adaptive network management, such as predicting pipe failure under climate extremes, or use agent-based modeling to resolve conflicts in multi-stakeholder irrigation systems (Ridha et al., 2022). Nevertheless, increasing model complexity also raises risks of over-parameterization and reduced transparency, particularly when system dynamics models are used to support policy or allocation decisions. Ensuring robustness, interpretability, and governance relevance remains a key unresolved challenge (Beven, 2006).

A critical challenge in agricultural water saving is the insufficient understanding of crops, farmland, and irrigation areas as a dynamically interconnected system. Traditional research often isolates water use efficiency traits, crop-microbe relationships, and the water-energy-food-ecosystem cycle, hindering the development of a holistic, systems-level perspective from molecular to regional scales. To overcome this limitation, a shift towards the system dynamics is essential (Zhou et al., 2025). Crucially, the role of system dynamics should be to integrate scale-appropriate representations—such as physiological efficiency metrics at plant scale, water accounting at field scale, and allocation constraints at regional scale—rather than enforcing mechanistic continuity across them (Perry, 2011). This necessitates the construction of multi-scale, multi-factor coupling models that explicitly represent the dynamic interactions between gene systems, crop-microbe systems, and irrigation water distribution systems (Fig. 9), while respecting physical and informational boundaries. Cross-scale integration theory has emerged as a frontier, unifying micro-scale mechanisms such as water transport in cells and crop growth with macro-scale hydrological processes through adaptive multi-scale models. By integrating remote sensing, field sensors, and numerical simulations, these models establish a full-chain framework of water-energy-solute-crop interactions. They quantify the conversion process from irrigation water to soil water, plant water, photosynthesis, biomass, and economic yield. The approach also proposes water-saving control theories to enhance efficiency at physiological, field, and regional levels. This theory bridges molecular-to-system behaviors, linking gene-editing-driven crop water use efficiency, microbial-mediated drought resistance, and smart irrigation optimization. It provides a unified basis for precision water management, resource allocation, and agricultural sustainability.

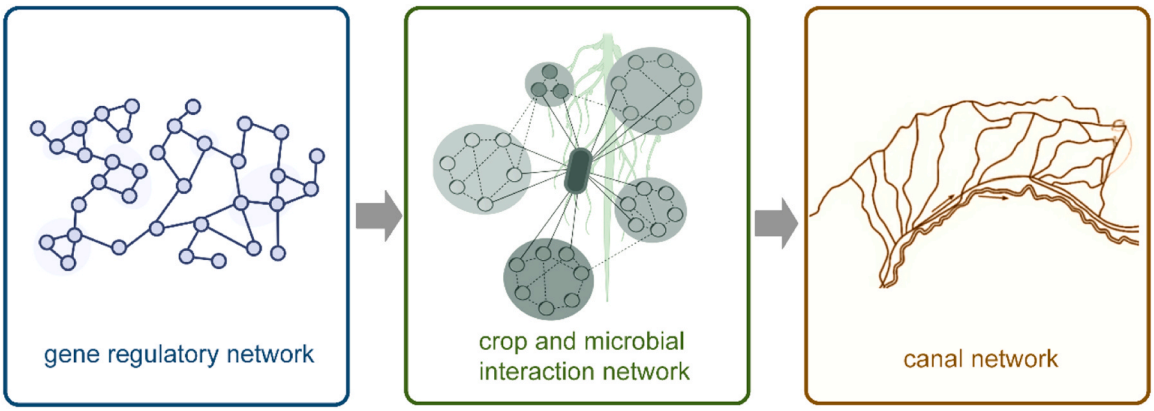


Fig. 9. | Hierarchical WUE system networks across operational scales. At the microscale, genetic regulatory networks govern crop responses to water availability. While synthetic biology and nanotechnology offer promising tools for enhancing WUE, a thorough understanding of their impact on crops at the genetic level – including potential unintended consequences – remains crucial. This requires detailed analysis of system dynamics within these regulatory networks. The mesoscale encompasses the complex interactions within the agricultural soil-crop-atmosphere-microbe system, where rhizospheric networks mediate water and nutrient uptake. Finally, the macroscale represents regional water conveyance and distribution networks, which determine water availability at the regional level. Achieving significant breakthroughs in WUE enhancement necessitates a holistic understanding of network dynamics and the implementation of targeted modulation strategies across these interconnected, multi-scale hierarchical systems.

4.5. AI-driven delineation and regulation of water resources in complex real-world systems

The precise regulation of water utilization hinges on a thorough understanding of crop water use and the intricate movement and transformation of water within agricultural water systems. With the progressive digitization of living organisms, farmlands, and irrigation districts into comprehensive data entities, the sustainable management of water resources within complex real-world ecosystems will fundamentally rely on artificial intelligence. This AI-driven paradigm shift is enabling a more nuanced and adaptive approach to water resource management across multiple scales (Fig. 10).

At the micro-scale, generative AI is revolutionizing the design of water-efficient technologies. This includes the discovery and development of advanced nanomaterials engineered to enhance water uptake and retention within plants, ultimately improving WUE and drought resistance (Chen et al., 2025; Rehman et al., 2024). Furthermore, AI-powered protein and genomic predictive modeling is dramatically

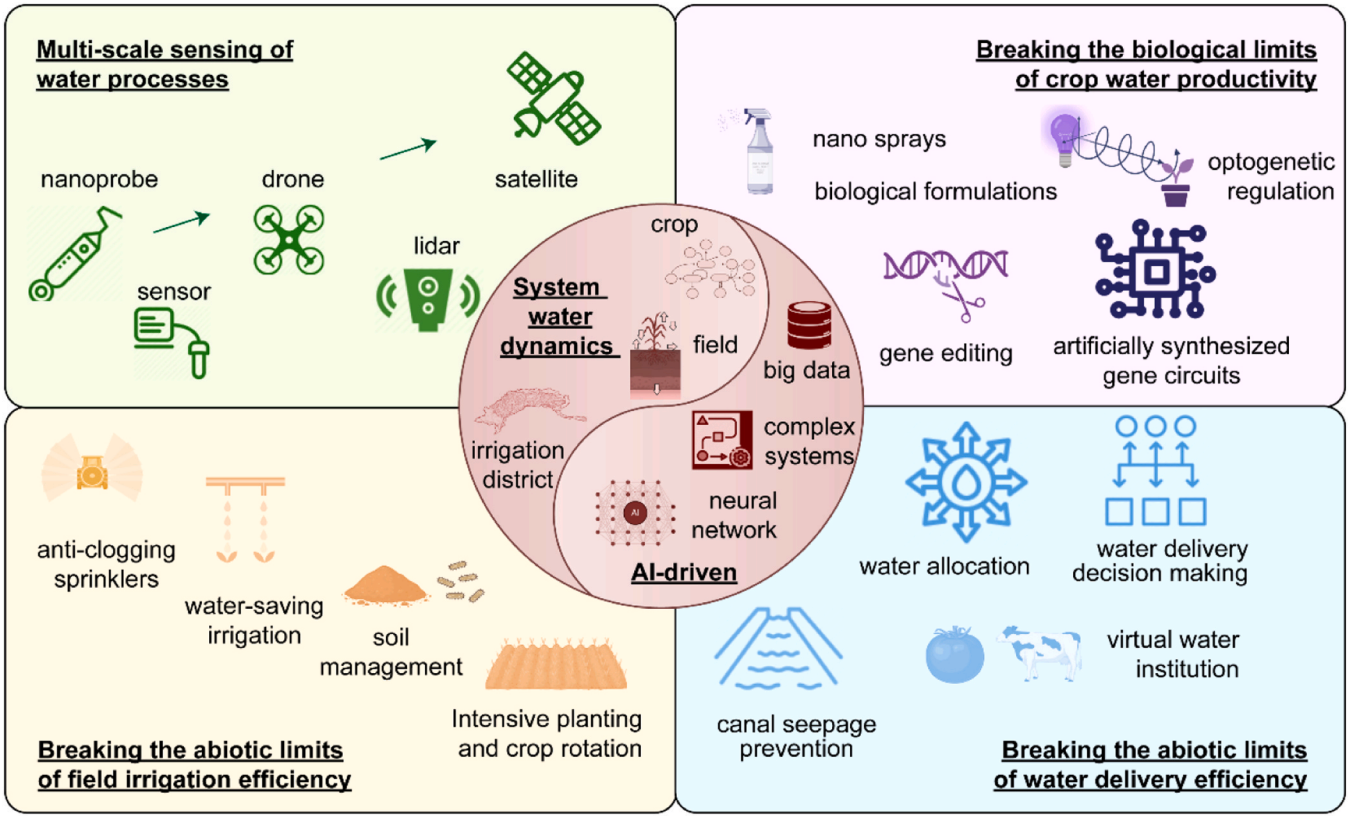


Fig. 10. | Future research directions in improving water use efficiency. Future research on WUE should be grounded in the dynamics of water systems. Aiming to break through biological and abiotic limits, the multi-scale sensing and AI-driven modeling paradigm should be adopted.

accelerating the breeding of drought-resistant crop varieties by optimizing plant genetics for water use efficiency (Le Roux et al., 2024; Xiong et al., 2025). These AI-driven innovations at the molecular level are crucial for minimizing water consumption and maximizing resource utilization (van Dijk et al., 2021). Moving to the meso-scale, AI is enabling high-fidelity simulations of complex soil-crop-atmosphere-microbe-water interactions. Machine learning algorithms are being trained on vast datasets of soil properties, crop phenology, meteorological data, and microbial activity to create predictive models of water flow, nutrient cycling, and plant growth (Aderele et al., 2025). These simulations, along with model interpretability, provide a deeper understanding of the intricate processes governing water use in agricultural systems, allowing for the development of targeted interventions to improve water management practices. At the macro-scale, AI is transforming water network optimization and water allocation decision-making. Reinforcement learning algorithms are being deployed to optimize the operation of complex canal networks, dynamically adjusting water distribution based on real-time conditions, predicted demand, and environmental constraints (Kåge et al., 2025). Furthermore, AI-powered decision support systems are integrating hydrological models, economic data, and stakeholder preferences to inform optimal water allocation strategies at the regional and national levels.

Importantly, while digital technologies and artificial intelligence substantially improve the accuracy of water demand estimation, system monitoring, and operational decision-making, these improvements do not automatically translate into actual agricultural water saving (Kilemo, 2022). The contribution of AI to real water saving critically depends on whether efficiency gains are coupled with explicit water allocation limits, consumptive use caps, or basin-scale management objectives. Without such constraints, enhanced operational efficiency may instead facilitate increased water use intensity or expanded irrigation activities.

Therefore, AI-driven water management should be understood as an enabling tool that supports agricultural water saving only when embedded within appropriate institutional, regulatory, and basin-scale governance frameworks. This holistic, AI-driven approach is essential for balancing competing water demands, minimizing environmental impacts, and ensuring equitable access to water resources (Molle and Tanouti, 2017). This holistic, AI-driven approach is essential for balancing competing water demands, minimizing environmental impacts, and ensuring equitable access to this vital resource.

5. Conclusions and future perspectives

Agricultural water-saving stands as a pivotal strategy for reconciling global food security with ecological sustainability. This study proposes a multi-scale framework encompassing molecular, field, and irrigation-district levels to address the dual challenges of breaking biological constraints and abiotic limits on water-use efficiency. At the molecular scale, breakthroughs in gene editing such as CRISPR-driven stomatal regulation, synthetic biology such as C_4 metabolic module transplantation, and nanomaterial-mediated delivery systems have decoded the genetic and metabolic determinants of water-use traits, enabling targeted manipulation of crop water dynamics. Field-scale innovations, including advanced water-saving irrigation equipment, integrated water and fertilizer management, soil improvement techniques, and a next-generation model, grounded in real-world environments, for predicting and regulating the dynamics of water, nutrients, crops, and ecosystems, have demonstrated measurable gains in both WUE and yield stability. At the regional scale, the integration of data on crop, atmosphere, soil, aquifer, and water conveyance and distribution systems has the potential to transform regional water allocation from static scheduling to real-time optimization. Crucially, the convergence of cross-scale sensing technologies, multi-omics and earth big data, and AI-assisted modeling and decision-making has bridged traditional disciplinary silos, offering unprecedented insights into the gene-to-landscape water continuum. As we decode the intricate dynamics of plants, farmlands, and irrigation districts, our capacity to regulate water utilization at corresponding scales will transcend current limitations.

Looking ahead, transformative progress will fundamentally depend on three critical frontiers. Firstly, breaking biological and abiotic limits is paramount; future efforts should focus on transcending the inherent physiological constraints of crops and the abiotic limitations of current management practices to achieve enhanced levels of water use efficiency. Achieving this will require overcoming significant challenges, including the complex influence of gene to water-use traits, and the difficulty of effectively implementing innovations in irrigation methods and equipment, as well as strategies for irrigation and water delivery, across diverse agricultural landscapes. Secondly, a comprehensive understanding of the complex, interconnected systems governing water use across molecular, field, and regional scales is essential. This requires elucidating the dynamic feedback loops and regulatory mechanisms that operate within and between these systems. The inherent complexity of these systems, coupled with data scarcity and uncertainties in climate projections, poses a considerable hurdle. Finally, leveraging a multi-scale perception and AI-driven paradigm will be vital. The integration of advanced multi-omics measurement, automated non-invasive filed monitoring, high spatiotemporal resolution regional observation with AI-powered analytics will be crucial for accurate reconstruction of the real environment, precise predictive modeling, and optimized resource allocation, facilitating the development of intelligent and adaptive water management strategies. Despite the promise of AI, challenges remain in data quality, model interpretability, and the potential for bias in algorithms. By strategically addressing these interconnected frontiers, we can unlock the full potential of agricultural water-saving technologies, laying the foundation for a comprehensive theory encompassing physiological water-saving, field management, and water allocation and delivery optimization, ultimately contributing to the sustainable utilization of global agricultural water resources.

CRediT authorship contribution statement

Xianzhi Deng: Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Conceptualization. **Liangsheng Shi:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Shaozhong Kang:** Supervision, Resources, Project administration, Conceptualization. **Zhiying Liu:** Writing – review & editing, Visualization, Data curation, Conceptualization. **Haolin Xu:** Writing – original draft, Visualization, Conceptualization. **Yujie Wang:** Writing – original draft, Visualization, Data curation, Conceptualization. **Zailin Huo:** Supervision, Resources, Project administration, Conceptualization. **Yunkai Li:** Supervision, Resources, Project administration, Conceptualization. **Jinmin Li:** Visualization, Supervision, Conceptualization. **Tinghan Wang:** Writing – original draft, Conceptualization.

Declaration of Competing Interest

No conflict of interest declared.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant No. 52425901). We would like to express our sincere gratitude to Professor Du Taisheng for his insightful discussions. Figures were created with BioRender.com, with permission.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.agwat.2026.110583.

Data availability

The supporting data for the literature collection can be found in the supplementary table.

References

Abbate, P.E., Dardanelli, J.L., Cantarero, M.G., Maturano, M., Melchiori, R.J.M., Suero, E.E., 2004. Climatic and water availability effects on water-use efficiency in wheat. Crop. Sci. 44, 474–483.

Abdulbaki, D., Al-Hindi, M., Yassine, A., Abou Najm, M., 2017. An optimization model for the allocation of water resources. J. Clean. Prod. 164, 994–1006.

Abdulla, F., Abdulla, C., Eslamian, S., 2021. Concept and technology of rainwater harvesting. Handbook of Water Harvesting and Conservation: Basic Concepts and Fundamentals, pp. 1–16.

Abioye, E.A., Abidin, M.S.Z., Mahmud, M.S.A., Buyamin, S., Ishak, M.H.I., Abd Rahman, M.K.I., Otuoze, A.O., Onotu, P., Ramli, M.S.A., 2020. A review on monitoring and advanced control strategies for precision irrigation. Comput. Electron. Agric. 173, 105441.

Aderele, M.O., Srivastava, A.K., Butterbach-Bahl, K., Rahimi, J., 2025. Integrating machine learning with agroecosystem modelling: Current state and future challenges. Eur. J. Agron. 168, 127610.

Aerts, N., Hickman, R., Van Dijken, A.J., Kaufmann, M., Snoek, B.L., Pieterse, C.M., Van Wees, S.C., 2024. Architecture and dynamics of the abscisic acid gene regulatory network. Plant. J. 119, 2538–2563.

Agyeman, B.T., Nouri, M., Appels, W.M., Liu, J., Shah, S.L., 2024. Learning-based multi-agent MPC for irrigation scheduling. Control Eng. Pract. 147, 105908.

Ahmar, S., Mahmood, T., Fiaz, S., Mora-Poblete, F., Shafique, M.S., Chattha, M.S., Jung, K.-H., 2021. Advantage of nanotechnology-based genome editing system and its application in crop improvement. Front. Plant Sci. 12, 663849.

Akhtar, N., Ilyas, N., Hayat, R., Yasmin, H., Noureldeen, A., Ahmad, P., 2021. Synergistic effects of plant growth promoting rhizobacteria and silicon dioxide nano-particles for amelioration of drought stress in wheat. Plant. Physiol. Biochem. 166, 160–176.

Al Mamun, M.R., Ahmed, A.K., Upoma, S.M., Haque, M.M., Ashik-E-Rabbani, M., 2025. IoT-enabled solar-powered smart irrigation for precision agriculture. Smart Agric. Technol. 10, 100773.

Alataway, A., Al-Ghobari, H., Mohammad, F., Dewidar, A., 2019. Lysimeter-based water use and crop coefficient of drip-irrigated potato in an arid environment. Agronomy 9, 756.

Amin, A.B., Rathnayake, K.N., Yim, W.C., Garcia, T.M., Wone, B., Cushman, J.C., Wone, B.W., 2019. Crassulacean acid metabolism abiotic stress-responsive transcription factors: a potential genetic engineering approach for improving crop tolerance to abiotic stress. Front. Plant Sci. 10, 129.

An, N., Miaomiao, R., Shan, J., Longbao, Y., Yanfei, W., Shuai, Z., 2025. Application and prospect of optical remote sensing technology in agricultural monitoring. J. Chin. Agric. Mech. 46, 199.

Andres, J., Blomeier, T., Zurbriggen, M.D., 2019. Synthetic switches and regulatory circuits in plants. Plant Physiol. 179, 862–884.

Ansari, R., Manna, A., Hazra, S., Bose, S., Chatterjee, A., Sen, P., 2024. Breeding 4.0 vis-à-vis application of artificial intelligence (AI) in crop improvement: an overview. N. Z. J. Crop. Hortic. Sci. 143.

Antoniuk, V., Manevski, K., Kørup, K., Larsen, R., Sandholt, I., Zhang, X., Andersen, M.N., 2021. Diurnal and seasonal mapping of water deficit index and evapotranspiration by an unmanned aerial system: A case study for winter wheat in Denmark. Remote. Sens. 13, 2998.

Antu, U.B., Islam, M.S., Ahmed, S., Arifuzzaman, M., Saha, S., Mitu, P.R., Sarkar, A.R., Mahiddin, N.A., Ismail, Z., Ibrahim, K.A., 2024. Emerging technologies for efficient water use in agriculture: A review of current trends and future directions. J. Water Process. Eng. 68, 106317.

Anzano, A., Bonanomi, G., Mazzoleni, S., Lanzotti, V., 2022. Plant metabolomics in biotic and abiotic stress: a critical overview. Phytochem. Rev. 21, 503–524.

Ashraf, M., Foolad, M.R., 2007. Roles of glycine betaine and proline in improving plant abiotic stress resistance. Environ. Exp. Bot. 59, 206–216.

Baldoni, E., Frugis, G., Martinelli, F., Benny, J., Paffetti, D., Buti, M., 2021. A comparative transcriptomic meta-analysis revealed conserved key genes and regulatory networks involved in drought tolerance in cereal crops. Int. J. Mol. Sci. 22, 13062.

Bandopadhyay, S., Li, X., Bowsher, A.W., Last, R.L., Shade, A., 2024. Disentangling plant- and environment-mediated drivers of active rhizosphere bacterial community dynamics during short-term drought. Nat. Commun. 15, 6347.

Barna, B., Gémes, K., Domoki, M., Bernula, D., Ferenc, G., Bálint, B., Nagy, I., Fehér, A., 2018. Arabidopsis NAP-related proteins (NRPs) contribute to the coordination of plant growth, developmental rate, and age-related pathogen resistance under short days. Plant. Sci. 267, 124–134.

Bastiaanssen, W.G., Bos, M.G., 1999. Irrigation performance indicators based on remotely sensed data: a review of literature. Irrig. Drain. Syst. 13, 291–311.

Bazrafshan, O., Zamani, H., Etedali, H.R., Moshizi, Z.G., Shamili, M., Ismaelpour, Y., Gholami, H., 2020. Improving water management in date palms using economic value of water footprint and virtual water trade concepts in Iran. Agric. Water Manag. 229, 105941.

Berger, M., Finkbeiner, M., 2010. Water footprinting: how to address water use in life cycle assessment? Sustainability 2, 919–944.

Beven, K., 2006. A manifesto for the equifinality thesis. J. Hydrol. 320, 18–36.

Blöschl, G., Sivapalan, M., 1995. Scale issues in hydrological modelling: A review. Hydrol. Process. 9, 251–290. https://doi.org/10.1002/hyp.3360090305.

Blum, A., 2009. Effective use of water (EUW) and not water-use efficiency (WUE) is the target of crop yield improvement under drought stress. Field Crops Res. 112, 119–123.

Boretti, A., Rosa, L., 2019. Reassessing the projections of the world water development report. NPJ Clean. Water 2, 15.

Borland, A.M., Hartwell, J., Weston, D.J., Schlauch, K.A., Tschaplinski, T.J., Tuskan, G. A., Yang, X., Cushman, J.C., 2014. Engineering crassulacean acid metabolism to improve water-use efficiency. Trends Plant. Sci. 19, 327–338.

Bosch, D.J., Eidman, V.R., Oosthuizen, L.K., 1987. A review of methods for evaluating the economic efficiency of irrigation. Agric. Water Manag. 12, 231–245.

Brink, S.C., 2021. Feed. World. Future Plant. Breed.

Burridge, J.D., Grondin, A., Vadez, V., 2022. Optimizing crop water use for drought and climate change adaptation requires a multi-scale approach. Front. Plant. Sci. 13, 824720.

Burt, C.M., Clemmens, A.J., Strelkoff, T.S., Solomon, K.H., Bliesner, R.D., Hardy, L.A., Howell, T.A., Eisenhauer, D.E., 1997. Irrigation performance measures: efficiency and uniformity. J. Irrig. Drain. Eng. 123, 423–442.

Bwambale, E., Abagale, F.K., Anornu, G.K., 2022. Smart irrigation monitoring and control strategies for improving water use efficiency in precision agriculture: A review. Agric. Water Manag. 260, 107324.

Bwambale, E., Abagale, F.K., Anornu, G.K., 2023. Data-driven modelling of soil moisture dynamics for smart irrigation scheduling. Smart Agric. Technol. 5, 100251.

Canone, D., Previati, M., Bevilacqua, I., Salvai, L., Ferraris, S., 2015. Field measurements based model for surface irrigation efficiency assessment. Agric. Water Manag. 156, 30–42.

Ceglar, A., Van der Wijngaart, R., De Wit, A., Lecerf, R., Boogaard, H., Seguini, L., Van den Berg, M., Toreti, A., Zampieri, M., Fumagalli, D., 2019. Improving WOFOST model to simulate winter wheat phenology in Europe: Evaluation and effects on yield. Agric. Syst. 168, 168–180.

Ceppi, A., Ravazzani, G., Corbari, C., Salerno, R., Meucci, S., Mancini, M., 2014. Real-time drought forecasting system for irrigation management. Hydrol. Earth Syst. Sci. 18, 3353–3366.

Chatrabhuj, Meshram, K., Mishra, U., Omar, P.J., 2024. Integration of remote sensing data and GIS technologies in river management system. Discov. Geosci. 2, 67.

Chatzimichael, K., Christopoulos, D., Stefanou, S., Tzouvelekas, V., 2020. Irrigation practices, water effectiveness and productivity measurement. Eur. Rev. Agric. Econ. 47 467498.

Chauhdary, J.N., Li, H., Jiang, Y., Pan, X., Hussain, Z., Javaid, M., Rizwan, M., 2023. Advances in Sprinkler Irrigation: A Review in the Context of Precision Irrigation for Crop Production. Agronomy 14, 47.

Chen, H., Song, Y., Wang, Y., Wang, H., Ding, Z., Fan, K., 2024. Zno nanoparticles: improving photosynthesis, shoot development, and phyllosphere microbiome composition in tea plants. J. Nanobiotechnol. 22, 389.

Chen, L., Huang, F., Liu, J., Yang, R., Hu, Q., Li, T., Zeng, Y., Dai, W., Qiu, T., White, J.C., 2025. Engineered Nanomaterials Enhance Crop Drought Resistance for Sustainable Agriculture. J. Agric. Food Chem. 73, 8715–8728.

Chen, Y., Leng, Y.-N., Zhu, F.-Y., Li, S.-E., Song, T., Zhang, J., 2023. Water-saving techniques: physiological responses and regulatory mechanisms of crops. Adv. Biotechnol. 1, 3.

Cheng, M., Wang, H., Fan, J., Zhang, S., Liao, Z., Zhang, F., Wang, Y., 2021. A global meta-analysis of yield and water use efficiency of crops, vegetables and fruits under full, deficit and alternate partial root-zone irrigation. Agric. Water Manag. 248, 106771.

Cheng, Q., Wei, T., Farbiak, L., Johnson, L.T., Dilliard, S.A., Siegwart, D.J., 2020. Selective organ targeting (SORT) nanoparticles for tissue-specific mRNA delivery and CRISPR–Cas gene editing. Nat. Nanotechnol. 15, 313–320.

Choudhary, S., Gaurav, V., Singh, A., Agarwal, S., 2019. Autonomous crop irrigation system using artificial intelligence. Int. J. Eng. Adv. Technol. 8, 46–51.

Clemmens, A.J., Bos, M.G., 1990. Statistical methods for irrigation system water delivery performance evaluation. Irrig. Drain. Syst. 4, 345–365.

Clemmens, A.J., Burt, C.M., 1997. Accuracy of irrigation efficiency estimates. J. Irrig. Drain. Eng. 123, 443–453.

Clothier, B., Jovanovic, N., Zhang, X., 2020. Reporting on water productivity and economic performance at the water-food nexus. Agric. Water Manag. 237, 106123.

Cominelli, E., Galbiati, M., Vavasseur, A., Conti, L., Sala, T., Vuylsteke, M., Leonhardt, N., Dellaporta, S.L., Tonelli, C., 2005. A guard-cell-specific MYB transcription factor regulates stomatal movements and plant drought tolerance. Curr. Biol. 15, 1196–1200.

Condon, A.G., 2020. Drying times: plant traits to improve crop water use efficiency and yield. J. Exp. Bot. 71, 2239–2252.

Condon, A.G., Richards, R.A., Rebetzke, G.J., Farquhar, G.D., 2002. Improving Intrinsic Water-Use Efficiency and Crop Yield. Crop. Sci. 42, 122–131. https://doi.org/10.2135/cropsci2002.1220.

Corbari, C., Salerno, R., Ceppi, A., Telesca, V., Mancini, M., 2019. Smart irrigation forecast using satellite LANDSAT data and meteo-hydrological modeling. Agric. Water Manag. 212, 283–294.

Crookston, B.S., Boren, D., Yost, M., Sullivan, T., Creech, E., Barker, B., Reid, C., 2025. Irrigation technology, irrigation dose, and crop genetic impacts on alfalfa yield and quality. Agric. Water Manag. 311, 109366.

Daszkowska-Golec, A., 2016. The role of abscisic acid in drought stress: how ABA helps plants to cope with drought stress. Drought Stress. Toler. Plants Vol. 2 Mol. Genet. Perspect. 123–151.

Debbarma, S., Bandyopadhyay, A., Bhadra, A., 2025. Streamflow Simulation Using a Hybrid Approach Combining HEC-HMS and LSTM Model in the Tlawng River Basin of Mizoram, India. Environ. Model. & Assess. 1–21.
Demirer, G.S., Zhang, H., Goh, N.S., Pinals, R.L., Chang, R., Landry, M.P., 2020. Carbon nanocarriers deliver siRNA to intact plant cells for efficient gene knockdown. Sci. Adv. 6, eaaz0495.
Deng, X., Zhang, Z., Hu, X., Li, J., Li, S., Su, C., Du, S., Shi, L., 2024. Estimation of photosynthetic parameters from hyperspectral images using optimal deep learning architecture. Comput. Electron. Agric. 216, 108540.
Demo, A.H., Asefa Bogale, G., 2024. Enhancing crop yield and conserving soil moisture through mulching practices in dryland agriculture. Front. Agron. 6, 1361697.
Deng, X., Hu, X., Shi, L., Su, C., Li, J., Du, S., Li, S., 2025. Deep learning-enabled exploration of global spectral features for photosynthetic capacity estimation. Front. Plant. Sci. 15, 1499875.
Deng, X., Shi, L., Wang, Y., Qiao, H., Wang, X., Ma, J., Zhang, Y., Zha, Y., Hu, X., 2026. Conserved and divergent gene regulatory networks for crop drought resistance. Nat. Commun.
Deng, X.-P., Shan, L., Zhang, H., Turner, N.C., 2006. Improving agricultural water use efficiency in arid and semiarid areas of China. Agric. Water Manag. 80, 23–40.
Deng, Z., Lv, M., Zhai, G., Cai, J., Gao, J., 2017. Research progress of new water-saving technology of surface irrigation in China. 2017 6th International Conference on Energy and Environmental Protection (ICEEP 2017). Atlantis Press, pp. 1268–1273.
Di Nardo, A., Di Natale, M., Giudicianni, C., Musmarra, D., Santonastaso, G.F., Simone, A., 2015. Water distribution system clustering and partitioning based on social network algorithms. Procedia Eng. 119, 196–205.
Ding, Y., Cheng, Y., Zhang, T., Ji, X., Qiao, R., Feng, H., 2021. Model. Dyn. Deep. Soil. Water root uptake maize mulched drip Irrig. Using. HYDRUS-2D.
Ding, Y., Zhu, Y., Feng, J., Zhang, P., Cheng, Z., 2020. Interpretable spatio-temporal attention LSTM model for flood forecasting. Neurocomputing 403, 348–359.
Dong, S., Jiang, Y., Dong, Y., Wang, L., Wang, W., Ma, Z., Yan, C., Ma, C., Liu, L., 2019. A study on soybean responses to drought stress and rehydration. Saudi J. Biol. Sci. 26, 2006–2017.
Dotaniya, M.L., Meena, V.D., Saha, J.K., Dotaniya, C.K., Mahmoud, A.E.D., Meena, B.L., Meena, M.D., Sanwal, R.C., Meena, R.S., Doutaniya, R.K., 2023. Reuse of poor-quality water for sustainable crop production in the changing scenario of climate. Environ. Dev. Sustain. 25, 7345–7376.
Dou, X., Yu, G., Chen, Z., Yang, M., Hao, T., Han, L., Liu, Z., Ma, L., Lin, Y., Zhu, X., 2024. High spatial variability in water use efficiency of terrestrial ecosystems throughout China is predominated by biological factors. Agric. For. Meteorol. 345, 109834.
Dubois, M., Claeys, H., Van den Broeck, L., Inzé, D., 2017. Time of day determines Arabidopsis transcriptome and growth dynamics under mild drought. Plant. Cell. & Environ. 40, 180–189.
Dunn, J., Hunt, L., Afsharinafar, M., Meselmani, M.A., Mitchell, A., Howells, R., Wallington, E., Fleming, A.J., Gray, J.E., 2019. Reduced stomatal density in bread wheat leads to increased water-use efficiency. J. Exp. Bot. 70, 4737–4748.
Ebmeyer, H., Hoffmann, C.M., 2022. Water use efficiency of sugar beet genotypes: A relationship between growth rates and water consumption. J. Agron. Crop. Sci. 208, 28–39.
Elkelish, A., Qari, S.H., Mazrou, Y.S., Abdelaal, K.A., Hafez, Y.M., Abu-Elsaoud, A.M., Batiha, G.E.-S., El-Esawi, M.A., El Nahhas, N., 2020. Exogenous ascorbic acid induced chilling tolerance in tomato plants through modulating metabolism, osmolytes, antioxidants, and transcriptional regulation of catalase and heat shock proteins. Plants 9, 431.
El-Shirbeny, M.A., Ali, A.M., Savin, I., Poddubskiy, A., Dokukin, P., 2021. Agricultural water monitoring for water management under pivot irrigation system using spatial techniques. Earth Syst. Environ. 5, 341–351.
Endo, A., Tsurita, I., Burnett, K., Orencio, P.M., 2017. A review of the current state of research on the water, energy, and food nexus. J. Hydrol. Reg. Stud. 11, 20–30.
Evans, R.G., King, B.A., 2010. Site-specific sprinkler irrigation in a water limited future. in: 5th National Decennial Irrigation Conference Proceedings, 5-8 December 2010, Phoenix Convention Center, Phoenix. American Society of Agricultural and Biological Engineers, Arizona USA, p. 1.
Evans, R.G., Sadler, E.J., 2008. Methods and technologies to improve efficiency of water use. Water Resour. Res. 44.
Falkenmark, M., 2020. Water resilience and human life support - global outlook for the next half century. Int. J. Water Resour. Dev. 36, 377–396. https://doi.org/10.1080/07900627.2019.1693983.
Falkenmark, M., Rockström, J., 2006. The New Blue and Green Water Paradigm: Breaking New Ground for Water Resources Planning and Management. J. Water Resour. Plann. Manag. 132, 129–132. https://doi.org/10.1061/(ASCE)0733-9496(2006)132:3(129).
Fan, Q., Long, X., Wang, X., Guo, W., 2024. Research and application of the method of measuring the effective utilization coefficient of irrigation water on farmland. in: Journal of Physics: Conference Series. IOP Publishing, 012023.
Feng, Z., 2011. Research on water-saving irrigation automatic control system based on internet of things, in: 2011. International Conference on Electric Information and Control Engineering. IEEE, pp. 2541–2544.
Fernández, J.E., Alcon, F., Diaz-Espejo, A., Hernandez-Santana, V., Cuevas, M.V., 2020. Water use indicators and economic analysis for on-farm irrigation decision: A case study of a super high density olive tree orchard. Agric. Water Manag. 237, 106074.
Ferreira, A., Rolim, J., Paredes, P., Cameira, M., do, R., 2023. Methodologies for water accounting at the collective irrigation system scale aiming at optimizing water productivity. Agronomy 13, 1938.
Filippi, P., Han, S.Y., Bishop, T.F.A., 2025. On crop yield modelling, predicting, and forecasting and addressing the common issues in published studies. Precis. Agric. 26, 8. https://doi.org/10.1007/s11119-024-10212-2.
Foley, D.J., Thenkabail, P.S., Aneece, I.P., Teluguntla, P.G., Oliphant, A.J., 2020. A meta-analysis of global crop water productivity of three leading world crops (wheat, corn, and rice) in the irrigated areas over three decades. Int. J. Digit. Earth 13, 939–975.
Freitas, R.S., da Silva, E.C., 2024. Unveiling the influence of stress memory: enhancing stress tolerance in seedlings through seed stage stress exposure. Plant Physiol. Rep. 29, 165–175.
Gago, J., Douthe, C., Florez-Sarasa, I., Escalona, J.M., Galmes, J., Fernie, A.R., Flexas, J., Medrano, H., 2014. Opportunities for improving leaf water use efficiency under climate change conditions. Plant. Sci. 226, 108–119.
Garcia-Caparros, P., Contreras, J.I., Baeza, R., Segura, M.L., Lao, M.T., 2017. Integral management of irrigation water in intensive horticultural systems of Almería. Sustainability 9, 2271.
Geilfus, C. -M., Zörb, C., Jones, J.J., Wimmer, M.A., Schmöckel, S.M., 2024. Water for agriculture: more crop per drop. Plant. Biol. J. 26, 499–507. https://doi.org/10.1111/plb.13652.
Geng, S., Liu, D., Xia, P., 2022. Research on China's water use efficiency of China based on a comparative analysis. Water Resour. Devel. Res. 22, 77–82.
Ghaffari, H., Tadayon, M.R., Nadeem, M., Razmjoo, J., Cheema, M., 2019. Foliage applications of jasmonic acid modulate the antioxidant defense under water deficit growth in sugar beet. Span. J. Agric. Res. 17, e0805–e0805.
Goap, A., Sharma, D., Shukla, A.K., Krishna, C.R., 2018. An IoT based smart irrigation management system using Machine learning and open source technologies. Comput. Electron. Agric. 155, 41–49.
Goche, T., Shargie, N.G., Cummins, I., Brown, A.P., Chivasa, S., Ngara, R., 2020. Comparative physiological and root proteome analyses of two sorghum varieties responding to water limitation. Sci. Rep. 10, 11835.
Goswami, P., Mathur, J., Srivastava, N., 2022. Silica nanoparticles as novel sustainable approach for plant growth and crop protection. Heliyon 8.
Grafton, R.Q., Williams, J., Perry, C.J., Molle, F., Ringler, C., Steduto, P., Udall, B., Wheeler, S.A., Wang, Y., Garrick, D., Allen, R.G., 2018. The paradox of irrigation efficiency. Science 361, 748–750. https://doi.org/10.1126/science.aat9314.
GU, S., FU, H., LI, J., 2003. Advances in real-time irrigation operation. Adv. Water Sci. 14, 660–666.
Guo, Y., Zhang, K., Jia, X., 2025. A Multilevel Analysis of Farmers' Adoption of Water-saving Irrigation Technology: Evidence From the North China Plain. Sage Open. 15, 21582440251316100. https://doi.org/10.1177/21582440251316100.
Haider, I., Raza, M.A.S., Iqbal, R., Aslam, M.U., Habib-ur-Rahman, M., Raja, S., Khan, M. T., Aslam, M.M., Waqas, M., Ahmad, S., 2020. Potential effects of biochar application on mitigating the drought stress implications on wheat (Triticum aestivum L.) under various growth stages. J. Saudi Chem. Soc. 24, 974–981.
Han, M., Wu, J., Wang, F., 2005. Redundancy theory and its application in agro-ecosystem management. Ying Yong Sheng tai xue bao= J. Appl. Ecol. 16, 375–378.
Hansana, P., Guo, X., Zhang, S., Kang, X., Li, S., 2023. Flood analysis using multi-scale remote sensing observations in Laos. Remote. Sens. 15, 3166.
Hatfield, J.L., Dold, C., 2019. Water-use efficiency: advances and challenges in a changing climate. Front. Plant. Sci. 10, 103.
Hatfield, J.L., Sauer, T.J., Prueger, J.H., 2001. Managing soils to achieve greater water use efficiency: a review. Agron. J. 93, 271–280.
He, L., Shi, L., Song, W., Shen, J., Wang, L., Hu, X., Zha, Y., 2024. Synergizing Intuitive Physics and Big Data in Deep Learning: Can We Obtain Process Insights While Maintaining State-Of-The-Art Hydrological Prediction Capability? Water Resour. Res. 60, e2024WR037582.
Hexiang, Z., Yujing, H., Rongxuan, B., Biyu, W., Jiabin, W., Delong, T., Haofang, Y., Beibei, H., 2025. Traceability simulation of drainage in irrigation areas along the south bank of the Yellow River based on the SWAT model. Front. Environ. Sci. 13, 1517836.
Hoekstra, A.Y., 2017. Water Footprint Assessment: Evolvement of a New Research Field. Water Resour. Manag. 31, 3061–3081. https://doi.org/10.1007/s11269-017-1618-5.
Hongren, S., Lingfa, M., Shuling, H., 2008. Effect of irrigation amount on water use efficiency and water consumption coefficient of alfalfa. Acta Agrestia Sin. 16.
Hoover, D.L., Abendroth, L.J., Browning, D.M., Saha, A., Snyder, K., Wagle, P., Witthaus, L., Baffaut, C., Biederman, J.A., Bosch, D.D., 2023. Indicators of water use efficiency across diverse agroecosystems and spatiotemporal scales. Sci. Total. Environ. 864, 160992.
Howell, T.A., 2001. Enhancing Water Use Efficiency in Irrigated Agriculture. Agron. J. 93, 281–289. https://doi.org/10.2134/agronj2001.932281x.
Howell, T.A., 2003. Irrigation efficiency. Encycl. Water Sci. 467, 500.
Hu, H.H., Kreymborg, L.R., Doeing, B.J., Baron, K.S., Jutila, S.A., 2006. Gridded snowmelt and rainfall-runoff CWMS hydrologic modeling of the Red River of the North Basin. J. Hydrol. Eng. 11, 91–100.
Hu, X., Shi, L., Lin, L., Li, S., Deng, X., Li, L., Bian, J., Lian, X., 2024. A novel hybrid modelling framework for GPP estimation: Integrating a multispectral surface reflectance based Vcmax25 simulator into the process-based model. Sci. Total. Environ. 921, 171182.
Huan, Z., Meng, L., 2021. Research on Application Effect of Construction Technology in Agricultural Water Conservancy Project. in: IOP Conference Series: Earth and Environmental Science. IOP Publishing, 042030.
Ikenberry, C.D., Soupir, M.L., Helmers, M.J., Crumpton, W.G., Arnold, J.G., Gassman, P. W., 2017. Simulation of daily flow pathways, tile-drain nitrate concentrations, and soil-nitrogen dynamics using SWAT. JAWRA J. Am. Water Resour. Assoc. 53, 1251–1266.
Islam, W., Idrees, A., Waheed, A., Zeng, F., 2022. Plant responses to drought stress: microRNAs in action. Environ. Res. 215, 114282.

Jägermeyr, J., Gerten, D., Heinke, J., Schaphoff, S., Kummu, M., Lucht, W., 2015. Water savings potentials of irrigation systems: dynamic global simulation. Hydrology & Earth System Sciences Discussions, p. 12.
Jat, M.L., Chandna, P., Gupta, R., Sharma, S.K., Gill, M.A., 2006. Laser land leveling: A precursor technology for resource conservation. Rice-Wheat Consort. Tech. Bull. Ser. 7, 48.
Javid, M.G., Sorooshzadeh, A., Moradi, F., Modarres Sanavy, S.A.M., Allahdadi, I., 2011. The role of phytohormones in alleviating salt stress in crop plants. Aust. j. crop sci. 5, 726–734.
Jensen, M.E., 2007. Beyond irrigation efficiency. Irrig. Sci. 25, 233–245.
Jha, U.C., Nayyar, H., Jha, R., Khurshid, M., Zhou, M., Mantri, N., Siddique, K.H., 2020. Long non-coding RNAs: emerging players regulating plant abiotic stress response and adaptation. BMC Plant. Biol. 20, 1–20.
Jia, X., Pang, X., Xu, M., Wang, C., Wei, H., Liu, J., Zhang, H., Li, D., Xu, X., Zhao, T., 2025. The Overexpression of SlPLATZ17 Can Increase the Tolerance of Tomatoes to Drought and Salt Stress. Int. J. Mol. Sci. 26, 1836.
Jiang, S., Liang, C., Cui, N., Zhao, L., Liu, C., Feng, Y., Hu, X., Gong, D., Zou, Q., 2020. Water use efficiency and its drivers in four typical agroecosystems based on flux tower measurements. Agric. For. Meteorol. 295, 108200.
Jin, D., Wei, Y., Zhang, K., 2025. The research on soil water utilization mechanism in drought-resistant crops based on big data. Adv. Resour. Res. 5, 329–349.
Jing, X., Liu, Y., Liu, X., Zhang, Yi, Wang, G., Yang, F., Zhang, Yani, Chang, D., Zhang, Z.-L., You, C.-X., 2024. Enhanced photosynthetic efficiency by nitrogen-doped carbon dots via plastoquinone-involved electron transfer in apple. Hortic. Res. 11, uhae016.
Kåge, L., Milić, V., Andersson, M., Wallén, M., 2025. Reinforcement learning applications in water resource management: a systematic literature review. Front. Water 7, 1537868.
Kaarthikeyan, G.M., Suresh, A., 2019. A study on understanding the adoption of water saving technology: A case study of drip irrigation. Int. J. Recent. Technol. Eng. 7, 1123–1130.
Kang, S., Zhang, L., Trout, T., 2017. Improving agricultural water productivity to ensure food security under changing environments. Agric. Water Manag. 179, 1–4.
Kang, Z., Lu, J., Zheng, S., Hu, X., Wang, L., Jiang, L., Zheng, Y., Lv, L., Gardea-Torresdey, J.L., White, J.C., 2025. Silica-Activated Redox Signaling Confers Rice with Enhanced Drought Resilience and Grain Yield. ACS nano.
Karagiannis, G., Tzouvelekas, V., Xepapadeas, A., 2003. Measuring irrigation water efficiency with a stochastic production frontier. Environ. Resour. Econ. 26, 57–72.
Karami, S., Shiran, B., Ravash, R., 2025. Molecular investigation of how drought stress affects chlorophyll metabolism and photosynthesis in leaves of C3 and C4 plant species: a transcriptome meta-analysis. Heliyon.
Karpatne, A., Jia, X., Kumar, V., 2024. Knowledge-guided machine learning: Current trends and future prospects. arXiv preprint arXiv:2403.15989.
Karavolias, N.G., Patel-Tupper, D., Gallegos Cruz, A., Litvak, L., Lieberman, S.E., Tjahjadi, M., Niyogi, K.K., Cho, M.-J., Staskawicz, B.J., 2024. Engineering quantitative stomatal trait variation and local adaptation potential by cis-regulatory editing. Plant Biotechnol. J. 22, 3442–3452.
Kashtoh, H., Baek, K.-H., 2021. Structural and functional insights into the role of guard cell ion channels in abiotic stress-induced stomatal closure. Plants 10, 2774.
Kausar, R., Wang, X., Komatsu, S., 2022. Crop proteomics under abiotic stress: from data to insights. Plants 11, 2877.
Kebede, H., Fisher, D.K., Sui, R., Reddy, K.N., 2014. Irrigation methods and scheduling in the Delta region of Mississippi: Current status and strategies to improve irrigation efficiency. Am. J. Plant. Sci. 5 2917.
Kelly, T.D., Foster, T., Schultz, D.M., 2024. Assessing the value of deep reinforcement learning for irrigation scheduling. Smart Agric. Technol. 7, 100403.
Khan, M.N., Khan, Z., Luo, T., Liu, J., Rizwan, M., Zhang, J., Xu, Z., Wu, H., Hu, L., 2020. Seed priming with gibberellic acid and melatonin in rapeseed: Consequences for improving yield and seed quality under drought and non-stress conditions. Ind. Crops Prod. 156, 112850.
Kilemo, D.B., 2022. The review of water use efficiency and water productivity metrics and their role in sustainable water resources management. Open. Access. Libr. J. 9, 1–21.
Kim, J., Mayorga-Burrezo, P., Song, S.-J., Mayorga-Martinez, C.C., Medina-Sánchez, M., Pané, S., Pumera, M., 2024. Advanced materials for micro/nanorobotics. Chem. Soc. Rev.
Kliebenstein, D.J., Monde, R.-A., Last, R.L., 1998. Superoxide dismutase in Arabidopsis: an eclectic enzyme family with disparate regulation and protein localization. Plant. Physiol. 118, 637–650.
Koech, R., Langat, P., 2018. Improving irrigation water use efficiency: A review of advances, challenges and opportunities in the Australian context. Water 10, 1771.
Kramer, P.J., 1932. The absorption of water by root systems of plants. Am. J. Bot. 148–164.
Kumar, K., Sridhar, J., Choudhary, V.K., Singh, H.K., Parameshwari, B., Kumar, K.S., Sahu, B., Dokka, N., Sivalingam, P.N., 2021. New innovations and approaches for biotic stress management of crops. in: Innovations in Agriculture for a Self-Reliant India. CRC Press, pp. 265–292.
López, M.-E., Roquis, D., Becker, C., Denoyes, B., Bucher, E., 2022. DNA methylation dynamics during stress response in woodland strawberry (Fragaria vesca). Hortic. Res. 9, uhac174.
Kumari, S., Kumar, S., Prakash, P., 2018. Exogenous application of cytokinin (6-BAP) ameliorates the adverse effect of combined drought and high temperature stress in wheat seedling. J. Pharmacogn. phytochem. 7, 1176–1180.
Le Roux, M.-S., Kunert, K.J., Cullis, C.A., Botha, A.-M., 2024. Unlocking wheat drought tolerance: the synergy of omics data and computational intelligence. Food Energy Secur. 13, e70024.
Leakey, A.D.B., Ferguson, J.N., Pignon, C.P., Wu, A., Jin, Z., Hammer, G.L., Lobell, D.B., 2019. Water Use Efficiency as a Constraint and Target for Improving the Resilience and Productivity of C_3 and C_4 Crops. Annu. Rev. Plant. Biol. 70, 781–808. https://doi.org/10.1146/annurev-arplant-042817-040305.
Lee, U., Chang, S., Putra, G.A., Kim, H., Kim, D.H., 2018. An automated, high-throughput plant phenotyping system using machine learning-based plant segmentation and image analysis. PloS one 13, e0196615.
Levidow, L., Zaccaria, D., Maia, R., Vivas, E., Todorovic, M., Scardigno, A., 2014. Improving water-efficient irrigation: Prospects and difficulties of innovative practices. Agric. Water Manag. 146, 84–94.
Li, B., Liu, D., Li, Q., Mao, X., Li, A., Wang, J., Chang, X., Jing, R., 2016. Overexpression of wheat gene TaMOR improves root system architecture and grain yield in Oryza sativa. J. Exp. Bot. 67, 4155–4167.
Li, G., Ma, D., 2025. Can a tiered water price policy improve the technical efficiency of crop irrigation for maize in the Heihe River Basin in Northwest China? Agric. Water Manag. 309, 109350.
Li, M.A., Jiahui, J., Yunfei, X., 2023. Operation and maintenance management of large-scale smart irrigation area based on BIM+ GIS technology. Mod. Agric. Sci. Technol. 23, 141–145.
Li, Q., Wei, M., Li, Y., Feng, G., Wang, Y., Li, S., Zhang, D., 2019. Effects of soil moisture on water transport, photosynthetic carbon gain and water use efficiency in tomato are influenced by evaporative demand. Agric. Water Manag. 226, 105818.
Li, R., Chang, Y., Wang, Z., 2021. Study of optimal allocation of water resources in Dujiangyan irrigation district of China based on an improved genetic algorithm. Water Supply 21, 2989–2999.
Li, T., Sun, M., Fu, Q., Cui, S., Liu, D., 2018. Analysis of irrigation canal system characteristics in Heilongjiang Province and the influence on irrigation water use efficiency. Water 10, 1101.
Liang, Z., Liu, X., Xiong, J., Xiao, J., 2020. Water allocation and integrative management of precision irrigation: A systematic review. Water 12, 3135.
Liang, Z., Liu, X., Xiao, J., Liu, C., 2021. Review of conceptual and systematic progress of precision irrigation. Int. J. Agric. Biol. Eng. 14, 20–31.
Lim, S.D., Lee, S., Choi, W.-G., Yim, W.C., Cushman, J.C., 2019. Laying the foundation for crassulacean acid metabolism (CAM) biodesign: expression of the C4 metabolism cycle genes of CAM in Arabidopsis. Front. Plant. Sci. 10, 101.
Lin, Y.-P., Chang, T.-K., Fan, C., Anthony, J., Petway, J.R., Lien, W.-Y., Liang, C.-P., Ho, Y.-F., 2017. Applications of information and communication technology for improvements of water and soil monitoring and assessments in agricultural areas—A case study in the taoyuan irrigation district. Environments 4, 6.
Liou, Y.-A., Kar, S.K., 2014. Evapotranspiration estimation with remote sensing and various surface energy balance algorithms—A review. Energies 7, 2821–2849.
Liu, J., Zhu, X., Yuan, S., Li, H., Tang, Y., 2022. Res. Dev. Trend Agric. Water-Sav. Sprink. Micro-Irrig. Equip. China.
Liu, X., Wei, Z., Hou, J., Wan, H., Zhang, Q., Ma, Y., Liu, F., 2022. Partial root-zone drying irrigation improves growth and physiology of tobacco amended with biochar by modulating phytohormonal profile and antioxidant system. Plant. Soil. 474, 561–579.
Loucks, D.P., Van Beek, E., 2017. Water resource systems planning and management: An introduction to methods, models, and applications. Springer.
Lu, J., Ookawa, T., Hirasawa, T., 2000. The effects of irrigation regimes on the water use, dry matter production and physiological responses of paddy rice. Plant. Soil. 223, 209–218.
Lu, Y., Liu, M., Li, C., Liu, X., Cao, C., Li, X., Kan, Z., 2022. Precision fertilization and irrigation: Progress and applications. AgriEngineering 4, 626–655.
MA, H., LUO, Z., CHEN, P., GUAN, B., 2019. Application of remote sensing technology in construction and management of irrigation and drainage engineering. HUBEI Agric. Sci. 58, 16.
Mai, Z., He, Y., Feng, C., Han, C., Shi, Y., Qi, W., 2024. Multi-objective modeling and optimization of water distribution for canal system considering irrigation coverage in artesian irrigation district. Agric. Water Manag. 301, 108959.
Mao, L.-L., Zhang, L.-Z., Zhang, S.-P., Evers, J.B., van der Werf, W., Wang, J.-J., Sun, H.-Q., Su, Z.-C., Spiertz, H., 2015. Resource use efficiency, ecological intensification and sustainability of intercropping systems. J. Integr. Agric. 14, 1542–1550.
Marsh, J.I., Hu, H., Gill, M., Batley, J., Edwards, D., 2021. Crop breeding for a changing climate: Integrating phenomics and genomics with bioinformatics. Theor. Appl. Genet. 134, 1677–1690.
Mastrangelo, A.M., Mazzucotelli, E., Guerra, D., De Vita, P., Cattivelli, L., 2011. Improvement of drought resistance in crops: from conventional breeding to genomic selection. In: Crop Stress and Its Management: Perspectives and Strategies. Springer, pp. 225–259.
Maurel, C., Nacry, P., 2020. Root architecture and hydraulics converge for acclimation to changing water availability. Nat. Plants 6, 744–749.
McShane, R.R. , Driscoll, K.P. , Sando, R. , 2017. A review of surface energy Liou and Kar, 2014; Lu et al., 2000; McShane et al., 2017; Yuan and Shen, 2013; Zhang et al., 2001; Mao et al., 2015; Tan et al., 2021 with remote sensing at high spatiotemporal resolution over large extents. Scientific Investigations Report.
Medrano, H., Flexas, J., Ribas-Carbó, M., Gulías, J., 2010. Measuring Water Use Efficiency in Grapevines. In: Delrot, S., Medrano, H., Or, E., Bavaresco, L., Grando, S. (Eds.), Methodologies and Results in Grapevine Research. Springer Netherlands, Dordrecht, pp. 123–134. https://doi.org/10.1007/978-90-481-9283-0_9.
Mengistu, T.D., Chung, I.-M., Kim, M.-G., Chang, S.W., Lee, J.E., 2022. Impacts and implications of land use land cover dynamics on groundwater recharge and surface runoff in East African Watershed. Water 14, 2068.
Miao, Q., Shi, H., Goncalves, J.M., Pereira, L.S., 2015. Field assessment of basin irrigation performance and water saving in Hetao, Yellow River basin: Issues to support irrigation systems modernisation. Biosyst. Eng. 136, 102–116.

Molden, D.J., Gates, T.K., 1990. Performance measures for evaluation of irrigation-water-delivery systems. J. Irrig. Drain. Eng. 116, 804–823.
Molle, F., Tanouti, O., 2017. Squaring the circle: Agricultural intensification vs. water conservation in Morocco. Agric. Water Manag. 192, 170–179.
Morianou, G., Karatzas, G.P., Arampatzis, G., Pisinaras, V., Kourgialas, N.N., 2025. Assessing Soil Water Dynamics in a Drip-Irrigated Grapefruit Orchard Using the HYDRUS 2D/3D Model: A Comparison of Unimodal and Bimodal Hydraulic Functions. Agronomy 15, 504.
Morison, J.I.L., Baker, N.R., Mullineaux, P.M., Davies, W.J., 2008. Improving water use in crop production. Philos. Trans. R. Soc. B Biol. Sci. 363, 639–658.
Mousavi, S.B., Sayfzadeh, S., Jabbari, H., Valadabadi, S.A., Masouleh, E.H., 2022. Effect of auxin foliar application on seed yield and fatty acids composition of two safflower genotypes under late-season drought. Plant Soil Environ. 68, 82.
Munjonji, L., Ayisi, K.K., 2021. Leaf gas exchange and δ13C in cowpea and triticale under water stress and well-watered conditions. Heliyon 7.
Mutema, M., Dhavu, K., 2022. Review of factors affecting canal water losses based on a meta-analysis of worldwide data. Irrig. Drain. 71, 559–573.
Nair, S., Johnson, J., Wang, C., 2013. Efficiency of irrigation water use: A review from the perspectives of multiple disciplines. Agron. J. 105, 351–363.
Nam, W.-H., Hong, E.-M., Choi, J.-Y., 2016. Assessment of water delivery efficiency in irrigation canals using performance indicators. Irrig. Sci. 34, 129–143.
Nawaz, A., Farooq, M., Ahmad, R., Basra, S.M.A., Lal, R., 2016. Seed priming improves stand establishment and productivity of no till wheat grown after direct seeded aerobic and transplanted flooded rice. Eur. J. Agron. 76, 130–137.
Nguyen, T.B.-A., Lefoulon, C., Nguyen, T.-H., Blatt, M.R., Carroll, W., 2023. Engineering stomata for enhanced carbon capture and water-use efficiency. Trends Plant Sci. 28, 1290–1309.
Ning, M., Li, Qinyan, Wang, Y., Li, Qingmao, Tao, Y., Zhang, F., Hu, F., Huang, L., 2025. Alternative splicing drives the functional diversification of a bHLH transcription factor in the control of growth and drought tolerance in rice. Sci. Bull. 70, 153–156.
Novoa, V., Rojas, O., Ahumada-Rudolph, R., Arumí, J.L., Munizaga, J., de la Barrera, F., Cabrera-Pardo, J.R., Rojas, C., 2023. Water footprint and virtual water flows from the Global South: Foundations for sustainable agriculture in periods of drought. Sci. Total. Environ. 869, 161526.
Orojloo, M., Shahdany, S.M.H., Roozbahani, A., 2018. Developing an integrated risk management framework for agricultural water conveyance and distribution systems within fuzzy decision making approaches. Sci. Total. Environ. 627, 1363–1376.
Oster, J.D., 1994. Irrigation with poor quality water. Agric. Water Manag. 25, 271–297.
Paltineanu, C., Septar, L., Moale, C., 2013. Crop water stress in peach orchards and relationships with soil moisture content in a Chernozem of Dobrogea. J. Irrig. Drain. Eng. 139, 20–25.
Pan, L., George-Jaeggli, B., Borrell, A., Jordan, D., Koller, F., Al-Salman, Y., Ghannoum, O., Cano, F.J., 2022. Coordination of stomata and vein patterns with leaf width underpins water-use efficiency in a C4 crop. Plant. Cell. & Environ. 45, 1612–1630.
Papacek, M., Christmann, A., Grill, E., 2019. Increased water use efficiency and water productivity of arabidopsis by abscisic acid receptors from Populus canescens. Ann. Bot. 124, 581–589.
Papanatsiou, M., Petersen, J., Henderson, L., Wang, Y., Christie, J.M., Blatt, M.R., 2019. Optogenetic manipulation of stomatal kinetics improves carbon assimilation, water use, and growth. Science 363, 1456–1459.
Parsegian, V.A., Rand, R.P., Rau, D.C., 1995. 3] Macromolecules and water: Probing with osmotic stress. in: Methods in Enzymology. Elsevier, pp. 43–94.
Patra, S.K., Poddar, R., Brestic, M., Acharjee, P.U., Bhattacharya, P., Sengupta, S., Pal, P., Bam, N., Biswas, B., Barek, V., 2022. Prospects of hydrogels in agriculture for enhancing crop and water productivity under water deficit condition. Int. J. Polym. Sci. 2022, 4914836.
Pereira, H., Marques, R.C., 2017. An analytical review of irrigation efficiency measured using deterministic and stochastic models. Agric. Water Manag. 184, 28–35.
Perry, C., 2011. Accounting for water use: Terminology and implications for saving water and increasing production. Agric. Water Manag. 98, 1840–1846.
Perry, C., Steduto, P., Allen, R.G., Burt, C.M., 2009. Increasing productivity in irrigated agriculture: Agronomic constraints and hydrological realities. Agric. water manag. 96, 1517–1524.
Pou, A., Medrano, H., Tomàs, M., Martorell, S., Ribas-Carbó, M., Flexas, J., 2012. Anisohydric behaviour in grapevines results in better performance under moderate water stress and recovery than isohydric behaviour. Plant. Soil. 359, 335–349.
Pratap, D., Kumar, S., Tripathi, U., Jayakkannan, L., 2024. Nanomaterial-induced modifications in plant physiology and genetics for optimal crop production strategies. Int. J. Environ. Clim. Change 14, 232–240.
Preite, L., Solari, F., Vignali, G., 2023. Technologies to optimize the water consumption in agriculture: A systematic review. Sustainability 15, 5975.
Preite, L., Solari, F., Vignali, G., 2024. Water Management Optimization in Agriculture: a Digital Model Development. Water Resour. Manag. 119.
Puppala, H., Ahuja, J., Tamvada, J.P., Peddinti, P.R., 2023. New technology adoption in rural areas of emerging economies: The case of rainwater harvesting systems in India. Technol. Forecast. Social. Change 196, 122832.
Qiu, M., Glass, Z., Chen, J., Haas, M., Jin, X., Zhao, X., Rui, X., Ye, Z., Li, Y., Zhang, F., 2021. Lipid nanoparticle-mediated codelivery of Cas9 mRNA and single-guide RNA achieves liver-specific in vivo genome editing of Angptl3. Proc. Natl. Acad. Sci. 118, e2020401118.
Rao, P.S., 1993. Rev. Sel. Lit. Indic. Irrig. Perform.
Raza, A., Salehi, H., Rahman, M.A., Zahid, Z., Madadkar Haghjou, M., Najafi-Kakavand, S., Charagh, S., Osman, H.S., Albaqami, M., Zhuang, Y., 2022. Plant hormones and neurotransmitter interactions mediate antioxidant defenses under induced oxidative stress in plants. Front. Plant. Sci. 13, 961872.
Raza, A., Charagh, S., Salehi, H., Abbas, S., Saeed, F., Poinern, G.E., Siddique, K.H., Varshney, R.K., 2023. Nano-enabled stress-smart agriculture: Can nanotechnology deliver drought and salinity-smart crops? J. Sustain. Agric. Environ. 2, 189–214.
Redillas, M.C., Jeong, J.S., Kim, Y.S., Jung, H., Bang, S.W., Choi, Y.D., Ha, S.-H., Reuzeau, C., Kim, J.-K., 2012. The overexpression of OsNAC9 alters the root architecture of rice plants enhancing drought resistance and grain yield under field conditions. Plant. Biotechnol. J. 10, 792–805.
Rehman, A., Khan, S., Sun, F., Peng, Z., Feng, K., Wang, N., Jia, Y., Pan, Z., He, S., Wang, L., 2024. Exploring the nano-wonders: unveiling the role of Nanoparticles in enhancing salinity and drought tolerance in plants. Front. Plant. Sci. 14, 1324176.
Ren, C., Wang, Y., Yu, L., Zhang, H., Xie, Z., 2023. An improved bi-level programming model for water resources allocation under multiple uncertainties. J. Environ. Manag. 345, 118882.
Ridha, T., Ross, A.D., Mostafavi, A., 2022. Climate change impacts on infrastructure: Flood risk perceptions and evaluations of water systems in coastal urban areas. Int. J. Disaster Risk Reduct. 73, 102883.
Rohwer, J.M., 2012. Kinetic modelling of plant metabolic pathways. J. Exp. Bot. 63, 2275–2292.
Ruggiero, A., Punzo, P., Landi, S., Costa, A., Van Oosten, M.J., Grillo, S., 2017. Improving plant water use efficiency through molecular genetics. Horticulturae 3, 31.
Sadras, V.O., Richards, R.A., 2014. Improvement of crop yield in dry environments: benchmarks, levels of organisation and the role of nitrogen. J. Exp. Bot. 65, 1981–1995.
Sadras, V.O., Grassini, P., Steduto, P., 2018. SOLAW TR07, 2011. Water (November).
Sallam, A., Alqudah, A.M., Dawood, M.F., Baenziger, P.S., Börner, A., 2019. Drought stress tolerance in wheat and barley: advances in physiology, breeding and genetics research. Int. J. Mol. Sci. 20, 3137.
Saltelli, A., Bammer, G., Bruno, I., Charters, E., Di Fiore, M., Didier, E., Nelson Espeland, W., Kay, J., Lo Piano, S., Mayo, D., 2020. Five ways to ensure that models serve society: a manifesto. Nature 582, 482–484.
Schmal, C., Reimann, P., Staiger, D., 2013. A circadian clock-regulated toggle switch explains AtGRP7 and AtGRP8 oscillations in Arabidopsis thaliana. PLoS Comput. Biol. 9, e1002986.
Seckler, D., Molden, D., Sakthivadivel, R., 2003. The concept of efficiency in water-resources management and policy. In: Kijne, J.W., Barker, R., Molden, D. (Eds.), Water Productivity in Agriculture: Limits and Opportunities for Improvement. CABI Publishing, UK, pp. 37–51. https://doi.org/10.1079/9780851996691.0037.
Seki, M., Umezawa, T., Urano, K., Shinozaki, K., 2007. Regulatory metabolic networks in drought stress responses. Curr. Opin. Plant. Biol. 10, 296–302.
Seleiman, M.F., Al-Suhaibani, N., Ali, N., Akmal, M., Alotaibi, M., Refay, Y., Dindaroglu, T., Abdul-Wajid, H.H., Battaglia, M.L., 2021. Drought stress impacts on plants and different approaches to alleviate its adverse effects. Plants 10, 259.
Semida, W.M., Abdelkhalik, A., Mohamed, G.F., Abd El-Mageed, T.A., Abd El-Mageed, S. A., Rady, M.M., Ali, E.F., 2021. Foliar application of zinc oxide nanoparticles promotes drought stress tolerance in eggplant (Solanum melongena L.). Plants 10, 421.
Serafeim, A.V., Fourniotis, N.T., Deidda, R., Kokosalakis, G., Langousis, A., 2024. Leakages in Water Distribution Networks: Estimation Methods, Influential Factors, and Mitigation Strategies—A Comprehensive Review. Water 16, 1534.
Shahriari, A., Omidi, H., Pakdehee, H.M., Khorzani, A.M., Ahmadi, K., 2019. Eff. Carbon Nanotub. seed priming germination Photosynth. Pigment. maize hybrids Drought Stress.
Shaikh, M., Birajdar, F., 2024. Groundwater and ecosystems: understanding the critical interplay for sustainability and conservation. EPRA Int. J. Multidiscip. Res. 10 181–186.
Shamshery, P., Winter, A.G., 2018. Shape and form optimization of on-line pressure-compensating drip emitters to achieve lower activation pressure. J. Mech. Des. 140, 035001.
Shao, H.-B., Li-Ye, C., Gang, W., Jin-Heng, Z., Zhao-Hua, L., 2007. Where is the road to bio-water-saving for the globe? Colloids Surf. B Biointerfaces 55, 251–255.
Sharafi, S., 2025. Enhanced seedling growth of annual medic under salt-drought stress through ultrasonic wave and magnetic field treatments. Appl. Water Sci. 15, 59. https://doi.org/10.1007/s13201-025-02399-0.
Sharma, N., Raman, H., Wheeler, D., Kalenahalli, Y., Sharma, R., 2023. Data-driven approaches to improve water-use efficiency and drought resistance in crop plants. Plant. Sci. 336, 111852.
Shinozaki, K., Yamaguchi-Shinozaki, K., 2007. Gene networks involved in drought stress response and tolerance. J. Exp. Bot. 58, 221–227.
Shuwen, W., Changli, Z., 2015. Study on farmland irrigation remote monitoring system based on ZigBee, in: 2015. International Conference on Computer and Computational Sciences (ICCCS). IEEE, pp. 193–197.
Singh, G., Sharma, D., Goap, A., Sehgal, S., Shukla, A.K., Kumar, S., 2019. Machine Learning based soil moisture prediction for Internet of Things based Smart Irrigation System, in: 2019. 5th International Conference on Signal Processing, Computing and Control (ISPCC). IEEE, pp. 175–180.
Singh, R., Kundu, D.K., Bandyopadhyay, K.K., 2010. Enhancing agricultural productivity through enhanced water use efficiency. J. Agric. Phys. 10, 1–15.
Sinha Roy, S., 2023. Myanmar Subnational Phone Surv. (MSPS) World Bank. Cover. Reliab. Represent.
Smith, R., Baillie, J., 2009. Defining precision irrigation: A new approach to irrigation management. In: Irrigation Australia 2009: Irrigation Australia Irrigation and Drainage Conference: Proceedings. University of Southern Queensland.
Sparks, D.L., 2012. Advances in agronomy. Academic Press.
Steduto, P., Hsiao, T.C., Fereres, E., Raes, D., 2012. Crop yield response to water. fao, Rome, Italy.
Stewart, B.A., Howell, T., 2003. Encyclopedia of Water Science (Print). CRC press.

Sun, J.L., Zhang, B., Zhang, S.J., 2014. Research on Structure Design of U type canal of seepage-proofing. Appl. Mech. Mater. 580 2033–2036.
Sun, Y., Oh, D.-H., Duan, L., Ramachandran, P., Ramirez, A., Bartlett, A., Tran, K.-N., Wang, G., Dassanayake, M., Dinneny, J.R., 2022. Divergence in the ABA gene regulatory network underlies differential growth control. Nat. Plants 8, 549–560.
Tabas, S.S., Samadi, S., 2022. Variational Bayesian dropout with a Gaussian prior for recurrent neural networks application in rainfall–runoff modeling. Environ. Res. Lett. 17, 065012.
Tambussi, E.A., Bort, J., Araus, J.L., 2007. Water use efficiency in C3 cereals under Mediterranean conditions: a review of physiological aspects. Ann. Appl. Biol. 150, 307–321.
Tan, Y., Hu, F., Chai, Q., Li, G., Zhao, C., Yu, A., Fan, H., Fan, Z., Yin, W., 2021. Optimizing water use between intercropped pea and maize through strip row ratio expansion and N fertilizer reduction in arid areas. Field Crops Res. 260, 108001.
Taotao, W., Bei, Z., Huihua, C., Jianwei, H.U.I., Long, H.A.O., Liang, C., 2024. Progress in developments and applications of the HYDRUS software and associated coupling model packages. Hydrogeol. & Eng. Geol.
Tekle, A.T., Alemu, M.A., 2016. Drought tolerance mechanisms in field crops. World J. Biol. Med. Sci. 3, 15–39.
Tian, X., Sarkis, J., Geng, Y., Qian, Y., Gao, C., Bleischwitz, R., Xu, Y., 2018. Evolution of China's water footprint and virtual water trade: A global trade assessment. Environ. Int. 121, 178–188.
Tom, T., Nirmala, M.M., Kumar, N.S., Reshmi, U.P., 2024. Unveiling the Agricultural Growth Catalyst: A Review on the Importance of Public Expenditure in the Agriculture Sector. In: El Khoury, R. (Ed.), Anticipating Future Business Trends: Navigating Artificial Intelligence Innovations, Studies in Systems, Decision and Control. Springer Nature Switzerland, Cham, pp. 243–256. https://doi.org/10.1007/978-3-031-63402-4_19.
Troch, P.A., Carrillo, G.A., Heidbüchel, I., Rajagopal, S., Switanek, M., Volkmann, T.H. M., Yaeger, M., 2009. Dealing with Landscape Heterogeneity in Watershed Hydrology: A Review of Recent Progress toward New Hydrological Theory. Geogr. Compass 3, 375–392. https://doi.org/10.1111/j.1749-8198.2008.00186.x.
Uga, Y., Sugimoto, K., Ogawa, S., Rane, J., Ishitani, M., Hara, N., Kitomi, Y., Inukai, Y., Ono, K., Kanno, N., 2013. Control of root system architecture by DEEPER ROOTING 1 increases rice yield under drought conditions. Nat. Genet. 45, 1097–1102.
Ullah, A., Manghwar, H., Shaban, M., Khan, A.H., Akbar, A., Ali, U., Ali, E., Fahad, S., 2018. Phytohormones enhanced drought tolerance in plants: a coping strategy. Environ. Sci. Pollut. Res. 25, 33103–33118.
Ullah, I., Hanping, M., Chuan, Z., Javed, Q., Azeem, A., 2017. Optimization of irrigation and nutrient concentration based on economic returns, substrate salt accumulation and water use efficiency for tomato in greenhouse. Arch. Agron. Soil. Sci. 63, 1748–1762.
Vadez, V., Pilloni, R., Grondin, A., Hajjarpoor, A., Belhouchette, H., Brouziyne, Y., Chehbouni, G., Kharrou, M.H., Zitouna-Chebbi, R., Mekki, I., 2023. Water use efficiency across scales: from genes to landscapes. J. Exp. Bot. 74, 4770–4788.
van Dijk, A.D.J., Kootstra, G., Kruijer, W., de Ridder, D., 2021. Machine learning in plant science and plant breeding. Iscience 24.
Vanuytrecht, E., Raes, D., Steduto, P., Hsiao, T.C., Fereres, E., Heng, L.K., Vila, M.G., Moreno, P.M., 2014. AquaCrop: FAO's crop water productivity and yield response model. Environ. Model. & Softw. 62, 351–360.
Velasco-Muñoz, J.F., Aznar-Sánchez, J.A., Belmonte-Ureña, L.J., Román-Sánchez, I.M., 2018. Sustainable water use in agriculture: A review of worldwide research. Sustainability 10, 1084.
Viani, F., Bertolli, M., Salucci, M., Polo, A., 2017. Low-cost wireless monitoring and decision support for water saving in agriculture. IEEE Sens. J. 17, 4299–4309.
Viets, F.G., 2015. Increasing Water Use Efficiency by Soil Management. In: Pierre, W.H., Kirkham, D., Pesek, J., Shaw, R. (Eds.), ASA, CSSA, and SSSA Books. American Society of Agronomy, Soil Science Society of America, Madison, WI, USA, pp. 259–274. https://doi.org/10.2134/1966.plantenvironment.c12.
Visser, M.A., Kumetat, G., Scott, G., 2024. Drought, water management, and agricultural livelihoods: Understanding human-ecological system management and livelihood strategies of farmer's in rural California. J. Rural. Stud. 109, 103339.
Wallace, J.S., 2000. Increasing agricultural water use efficiency to meet future food production. Agric. Ecosyst. & Environ. 82, 105–119.
Wang, J., Sha, J., Liu, R., Ren, S., Zhao, X., Liu, G., 2024. The Study of Various Regression Models Establishment to Identify Farmland Soil Moisture Content at Different Depths Using Unmanned Aerial Vehicle Multispectral Data: A Case in North China Plain. Water 16, 807.
Wang, L., Liu, J., Yang, L., Chen, Z., Wang, X., Ouyang, B., 2013. Applications of unmanned aerial vehicle images on agricultural remote sensing monitoring. Trans. Chin. Soc. Agric. Eng. 29, 136–145.
Wang, M., Ellsworth, P.Z., Zhou, J., Cousins, A.B., Sankaran, S., 2016. Evaluation of water-use efficiency in foxtail millet (Setaria italica) using visible-near infrared and thermal spectral sensing techniques. Talanta 152, 531–539.
Wang, P., Wang, W., Rong, Y., Zhang, C., Wang, C., Wang, X., Huo, Z., 2025. A novel conceptual model coupling crop growth and soil water-heat-salt processes in arid area. J. Hydrol. 647, 132284.
Wang, Q., Xu, J., Pu, X., Lv, H., Liu, Y., Ma, H., Wu, F., Wang, Qingjun, Feng, X., Liu, T., 2021. Maize DNA methylation in response to drought stress is involved in target gene expression and alternative splicing. Int. J. Mol. Sci. 22, 8285.
Wang, Yanling, Shi, L., Hu, X., Song, W., Wang, L., 2023. Multiphysics-Informed Neural Networks for Coupled Soil Hydrothermal Modeling. Water Resour. Res. 59, e2022WR031960. https://doi.org/10.1029/2022WR031960.
WANG, Y., DONG, B., QIAO, Y., YANG, H., JIN, L., LIU, J., LIU, M., 2019. Experimental study on soil water threshold of luxury transpiration in winter wheat leaves during flowering and filling stage. Chin. J. Eco-Agric. 27, 1024–1032.
Ward, F.A., Pulido-Velazquez, M., 2008. Water conservation in irrigation can increase water use. Proc. Natl. Acad. Sci. U. S. A. 105, 18215–18220. https://doi.org/10.1073/pnas.0805554105.
Wei, S., Xu, T., 2025. An LSTM approach to deciphering irrigation operations from remote sensing and groundwater levels records. Agric. Water Manag. 308, 109273.
Wenli, S., Shahrajabian, M.H., Huang, Q., 2020. Soybean seeds treated with single walled carbon nanotubes (SwCNTs) showed enhanced drought tolerance during germination. Int. J. Adv. Biol. Biomed. Res. 8, 9–16.
Wu, H., Zhang, Y., Zhang, W., Pei, X., Zhang, C., Jia, S., Li, W., 2015. Transcriptomic analysis of the primary roots of Alhagi sparsifolia in response to water stress. PLoS. One 10, e0120791.
Wu, J., Ma, C., Li, F., 2022. Microbial community structure and function in paddy soil as affected by water-saving irrigation mode. Eur. J. Soil. Biol. 113, 103450.
Wu, X., Zhou, H., 2011. Influence factor to water transport index in canal system in Xinjiang Irrigation area. Yellow River 9, 93–97.
Wu-quan, H.E., Bin, Y.A.O., Di, W.U., Yu-bao, W., Qian-xi, L.I., 2024. The Development History and Prospects of Channel Seepage Prevention Engineering Technology in New China. China Rural Water & Hydropower.
Xiao, J., Zhou, Y., Wang, H., Qiao, Y., Zhao, P., Cao, Y., Liu, X., Yang, Y., Lin, X., Xu, S., 2024. Integrative Omics reveals genetic basis and TaMYB7-A1's function in wheat WUE and drought resilience. bioRxiv 2024 12 (12), 628287.
Xiong, Y., Wang, R., Yang, Q., 2025. Artificial intelligence empowers maize drought resistance breeding: Current status, technological breakthroughs, and future prospects. Adv. Resour. Res. 5, 817–833.
Xu, L., Naylor, D., Dong, Z., Simmons, T., Pierroz, G., Hixson, K.K., Kim, Y.-M., Zink, E. M., Engbrecht, K.M., Wang, Y.I., 2018. Drought delays development of the sorghum root microbiome and enriches for monoderm bacteria. Proc. Natl. Acad. Sci. 115, E4284–E4293.
Xu, L., Dong, Z., Chiniquy, D., Pierroz, G., Deng, S., Gao, C., Diamond, S., Simmons, T., Wipf, H.M.-L., Caddell, D., 2021. Genome-resolved metagenomics reveals role of iron metabolism in drought-induced rhizosphere microbiome dynamics. Nat. Commun. 12, 3209.
Xu, Qinhong, Boelens, R., Veldwisch, G.J., 2024. High-Efficiency Irrigation. Int. J. Commons 18, 148–163.
Xu, Y., Zhang, X., Li, H., Zheng, H., Zhang, J., Olsen, M.S., Varshney, R.K., Prasanna, B. M., Qian, Q., 2022. Smart breeding driven by big data, artificial intelligence, and integrated genomic-enviromic prediction. Mol. Plant. 15, 1664–1695.
Yan, R., Zhao, N., Wang, Y., Liu, X., 2024. The impact of water rights trading on water resource use efficiency: Evidence from China's water rights trading pilots. Water Resour. Econ. 46, 100241.
Yan, S., Weng, B., Jing, L., Bi, W., Yan, D., 2022. Adaptive pathway of summer maize under drought stress: Transformation of root morphology and water absorption law. Front. Earth Sci. 10, 1020553.
Yang, F., Zhu, L., Liu, Z., Zhou, J., 2025. ShCAS1-synthesized β-Caryophyllene enhances broad-spectrum stress resistance in Solanum lycopersicum and Solanum habrochaites. Plant. Physiol. Biochem., 109726
Yang, J., Ordiz, M.I., Jaworski, J.G., Beachy, R.N., 2011. Induced accumulation of cuticular waxes enhances drought tolerance in Arabidopsis by changes in development of stomata. Plant. Physiol. Biochem. 49, 1448–1455.
Yang, Q., Shi, L., Han, J., Zha, Y., Zhu, P., 2019. Deep convolutional neural networks for rice grain yield estimation at the ripening stage using UAV-based remotely sensed images. Field Crops Res. 235, 142–153.
Yang, S., Vanderbeld, B., Wan, J., Huang, Y., 2010. Narrowing down the targets: towards successful genetic engineering of drought-tolerant crops. Mol. Plant. 3, 469–490.
Yang, X., Cushman, J.C., Borland, A.M., Edwards, E.J., Wullschleger, S.D., Tuskan, G.A., Owen, N.A., Griffiths, H., Smith, J.A.C., De Paoli, H.C., 2015. A roadmap for research on crassulacean acid metabolism (CAM) to enhance sustainable food and bioenergy production in a hotter, drier world. New. Phytol. 207, 491–504.
Yin, R., Xia, K., Xu, X., 2023. Spatial transcriptomics drives a new era in plant research. Plant. J. 116, 1571–1581.
Yousaf, W., Awan, W.K., Kamran, M., Ahmad, S.R., Bodla, H.U., Riaz, M., Umar, M., Chohan, K., 2021. A paradigm of GIS and remote sensing for crop water deficit assessment in near real time to improve irrigation distribution plan. Agric. Water Manag. 243, 106443.
Yuan, C., Pan, Y., Jing, S., 2025. Simulation soil water-salt dynamic and groundwater depth of spring maize based on SWAP model in salinized irrigation district. Comput. Electron. Agric. 231, 109992.
Yuan, F., Yang, H., Xue, Y., Kong, D., Ye, R., Li, C., Zhang, J., Theprungsirikul, L., Shrift, T., Krichilsky, B., 2014. OSCA1 mediates osmotic-stress-evoked Ca2+ increases vital for osmosensing in Arabidopsis. Nature 514, 367–371.
Yuan, K., Yang, Z., Wang, S., 2021. Water scarcity and adoption of water-saving irrigation technologies in groundwater over-exploited areas in the North China Plain. Irrig. Sci. 39, 397–408. https://doi.org/10.1007/s00271-021-00726-2.
Yuan, Y., Lee, H., Hu, H., Scheben, A., Edwards, D., 2018. Single-cell genomic analysis in plants. Genes 9, 50.
Yuan, Y., Xi, C., Jing, Q., 2019. Advance in agricultural drought monitoring using remote sensing data. Spectrosc. Spectr. Anal. 39, 1005–1012.
Yuan, Z., Shen, Y., 2013. Estimation of agricultural water consumption from meteorological and yield data: a case study of Hebei, North China. PloS. One 8, e58685.
Zablotskii, V., Gorobets, O., Gorobets, S., Polyakova, T., 2025. Effects of Static and Low-Frequency Magnetic Fields on Gene Expression. Magn. Reson. Imaging 62, 317–334. https://doi.org/10.1002/jmri.29726.
Zahoor, S.A., Ahmad, S., Ahmad, A., Wajid, A., Khaliq, T., Mubeen, M., Hussain, S., Din, M.S.U., Amin, A., Awais, M., 2019. Improving water use efficiency in agronomic crop production. Agron. Crop. Vol. 2 Manag. Pract. 13–29.

Zhang, D., Guo, P., 2016. Integrated agriculture water management optimization model for water saving potential analysis. Agric. Water Manag. 170, 5–19.
Zhang, G., Hou, Y., Zhang, H., Fan, H., Wen, X., Han, J., Liao, Y., 2022. Optimizing planting pattern and nitrogen application rate improves grain yield and water use efficiency for rain-fed spring maize by promoting root growth and reducing redundant root growth. Soil. Tillage Res. 220, 105385.
Zhang, L., Dawes, W.R., Walker, G.R., 2001. Response of mean annual evapotranspiration to vegetation changes at catchment scale. Water Resour. Res. 37, 701–708.
Zhang, P., Yang, H., Liu, F., Li, X., 2023. Stress memories for better tolerance in plants—A potential strategy for crop breeding. Agronomy. MDPI.
Zhang, S., Fang, G., Zhang, D., Ju, M., Zhong, H., 2023. Water resources optimal allocation model for coordinating regional multi-level water resources managers' interests. Front. Environ. Sci. 11, 1152296.
Zhang, Z., Shao, H., Xu, P., Chu, L., Lu, Z., Tian, J., 2007. On evolution and perspectives of bio-watersaving. Colloids Surf. B Biointerfaces 55, 1–9.
Zhao, C., Zhang, Y., Du, J., Guo, X., Wen, W., Gu, S., Wang, J., Fan, J., 2019. Crop phenomics: current status and perspectives. Front. Plant. Sci. 10, 714.
Zheng, H., Mei, P., Wang, W., Yin, Y., Li, H., Zheng, M., Ou, X., Cui, Z., 2023. Effects of super absorbent polymer on crop yield, water productivity and soil properties: A global meta-analysis. Agric. Water Manag. 282, 108290.
Zhou, X., Sheng, Z., Manevski, K., Zhao, R., Zhang, Q., Yang, Yanmin, Han, S., Liu, J., Yang, Yonghui, 2025. Can system dynamics explain long-term hydrological behaviors? The role of endogenous linking structure. Hydrol. Earth Syst. Sci. 29, 159–177.
Zhu, J.-K., 2001. Plant salt tolerance. Trends Plant. Sci. 6, 66–71.
Zhu, X., Xiong, L., 2013. Putative megaenzyme DWA1 plays essential roles in drought resistance by regulating stress-induced wax deposition in rice. Proc. Natl. Acad. Sci. 110, 17790–17795.
Zinkernagel, J., Maestre-Valero, J.F., Seresti, S.Y., Intrigliolo, D.S., 2020. New technologies and practical approaches to improve irrigation management of open field vegetable crops. Agric. Water Manag. 242, 106404.
Zwart, S.J., Bastiaanssen, W.G., 2004. Review of measured crop water productivity values for irrigated wheat, rice, cotton and maize. Agric. Water Manag. 69, 115–133.